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Review article

Research progress on bioleaching recovery technology of spent lithium-ion batteries

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ABSTRACT

Bioleaching of lithium-ion batteries is a microbially catalyzed process. Under the action of redox, acid leaching and complexation in the presence of microorganisms, the valuable metals in the cathode material enter the liquid phase as ions and are subsequently recovered from the succeeding process. This technique has the advantages of being inexpensive, environmentally friendly and having simple needs. However, it is still in development and has not yet commercialized. In this paper, the technology is fully discussed based on numerous excellent studies. The contents include commonly utilized microorganisms, bioleaching mechanism, microbial stress response and metabolic activation, enhancement strategies, leaching characteristics and interfacial phenomena, process evaluation, and a critical discussion of recent research breakthroughs. They give readers with comprehensive and in-depth understanding on the bioleaching of lithium-ion batteries and help to improve the technology's industrialization. Researchers can make new explorations from the potential research directions and methods presented in this work to make biotechnology better serve resource recovery and social development.

1. Introduction

Lithium-ion batteries (LIBs) have the advantages of small size, high energy density, and no memory effect, and their structure is shown in Fig. 1. It was first commercially produced by Sony company in 1991 and is now widely used in cell phones, cameras, laptops, electric vehicles, and power grids (Sethurajan and Gaydardzhiev, 2021; Li et al., 2018; Chen et al., 2021). Driven by energy technology changes and emerging technologies, especially new energy vehicles entered a period of rapid development in 2011, the global output of LIBs has increased exponentially. The global LIBs output in 2022 will be close to 1000 GWh. China accounts for more than half of total global LIBs production in the long term, with 750 GWh in 2022, a year-over-year rise of more than 130% (Lai et al., 2021).

LIBs have a normal lifespan of 5–10 years and will thus become a major metal waste stream in the coming years (Yu et al., 2021a, 2021b). As shown in Fig. 2, the vast number of spent LIBs that are soon to emerge may cause serious social problems if they are not well-placed. On the one hand, when LIBs are piled up or landfilled, toxic substances can seep into the soil and groundwater, causing environmental damage and posing a health risk (Naseri et al., 2019a; Iturrondobeitia et al., 2022; Yu et al., 2022). On the other hand, LIBs contain a large amount of valuable metals, far more than their content in primary mineral resources. Li, Ni, Co, and Mn account for about 5%–7%, 5%–10%, 5%–20%, and 5%–15%, respectively, with an additional 5%–10% of other base metals (Cu, Fe, and Al) (Xu et al., 2022; Yang et al., 2022). In the next 20 years, market demand for lithium is expected to exceed six times the capacity of its mineral resource (Zheng et al., 2017). Therefore, effective LIBs recycling is both reasonable and necessary.

Currently, the main recovery methods for LIBs are mechanical crushing-sorting, pyrometallurgy, hydrometallurgy, bioleaching and materials regeneration, and their characteristics are listed in Table 1 (Sun and Qiu, 2012; Lee et al., 2022; Ghassa et al., 2021; Yan et al., 2022; Swain, 2017; Zhao et al., 2020). Mechanical crushing-sorting is usually used as the front-end pretreatment technology for the recovery process, and pyrometallurgical and hydrometallurgical technologies have been applied in commercial production (Chagnes and Pospiech, 2013; Golmohammadzadeh et al., 2017). In pyrometallurgical processes, metals such as cobalt and nickel form polymetallic alloys at high temperatures. Georgi-Maschler et al. (2012) conducted reductive melting of LiNi$_x$Coy$_M$n1$_{-x-y}$O2 (NCM) materials at 800 °C to produce Fe–Co–Ni–Mn alloys and Li-bearing slag, the latter of which was further

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leached with sulfuric acid to recover Li. In hydrometallurgical processes, the "H_2SO_4+H_2O_2" leaching system has demonstrated to be effective, recovering more than 95% of the metals from the cathodic material (Shin et al., 2005; Zheng et al., 2018a). Yang et al. (2020) leached NCM materials at a H_2O_2 volume concentration of 3%, a sulfuric acid concentration of 3.0 mol/L, and 80 °C. The leaching rates of Li, Ni, Co, and Mn were 97.8%, 98.1%, 96.5%, and 97.0%, respectively. However, pyrometallurgy requires high temperature conditions (generally greater than 500 °C) while causing lithium loss. High acid concentrations (3–6 mol/L) are required in hydrometallurgy, leading to equipment corrosion and safety hazards (Yao et al., 2018; Li et al., 2023; Chen et al., 2011). A deep eutectic solvent composed of choline chloride and ethylene glycol has been proposed with cobalt solubility greater than 90%, but the technique still requires a high temperature (180 °C) (Tran et al., 2019). In the new situation, where society pays more attention to production costs, environmental protection, operational safety, and secondary pollution, pyrometallurgical and hydrometallurgical technologies are under severe limitation. Therefore, people have turned their attention to low-cost, low-energy, and low-pollution bioleaching technology. It is estimated that the investment cost required for the bioleaching method is 1/3 to 1/2 that of the conventional method (Gao et al., 2017; Heydarian et al., 2018; Roy et al., 2022).

As an viable recycling technology for LIBs, bioleaching has not yet been used in commercial production. Therefore, this paper reviews the recent progress of bioleaching technology for spent LIBs, with a focus on bioleaching mechanism and enhancement measures. Our effort intends to improve the in-depth understanding of bioleaching technology, thus providing references for related scholars and companies to promote its development and industrial application.

2. Pretreatment of LIBs

To obtain the cathodic material to be leached, a pretreatment process of "Discharge-Crushing-Separation-Dissolution/Pyrolysis-Fine grinding-Sieving" as depicted in Fig. 3 is usually adopted (Yang et al., 2020; Li et al., 2010; Chandran et al., 2021). To avoid safety hazards, spent LIBs need to be thoroughly discharged first. Physical discharge method was found to be unstable and unsuitable for large-scale production. Soaking the battery in a 0.8 mol/L NaCl solution for 3 h is the best scheme for chemical discharge, which can reduce the voltage to below 0.5 V (Yao et al., 2020). Crushing and separation are used to separate the electrode material from other substances (organic matter, etc.) to achieve enrichment of the electrode material. Zhang et al. (2021) suggested that the mechanical method is highly productive and the most promising technology for industrialization. The separated cathode is cut into small pieces and put into a chemical solution (e.g., N-methyl-2-pyrrolidone, NMP) and stirred to dissolve the organic binder to obtain the electrode material. Alternatively, pyrolysis is used to remove insoluble organic residues from the electrode material's surface. High-temperature vacuum pyrolysis can decompose organic matter into small molecule liquids or gases, but the method is demanding in terms of equipment, complex in operation, and not easily industrialised (Sun and Qiu, 2011). The purer cathode material obtained by the above process is dried, finely ground, and sieved to obtain a cathode black powder suitable for biorecycling.

3. Bioleaching of cathode materials

3.1. Microorganism

Most of the microorganisms involved in LIBs bioleaching are acidophilic autotrophic bacteria, such as *Acidithiobacillus ferrooxidans* (*A. ferrooxidans)*, which are commonly found in acidic pit water and soil. Some mixotrophic or heterotrophic microorganisms, such as *Sulfobacillus thermosulfidooxidans (S. thermosulfidooxidans)* and *Ferroplasma* spp. also have related bioleaching functions (Li et al., 2022). *Aspergillus niger* (*A. niger)* is a fungus that can be easily isolated from substances such as soil. It obtains energy by breaking down metal-containing compounds into basic elements, thereby secreting biological acids such as citric, oxalic, and gluconic acids (Bahaloo-Horeh et al., 2018). A summary of the growth and functional properties of these microorganisms are highlighted in Table 2. Other microorganisms like *Alicyclobacillus*. spp. also exist in some microbial communities studied (Wu et al., 2020; Niu et al., 2014).

From Table 2, most of the microorganisms are acidophilic and mesophilic, with ideal temperatures and pH ranges of 25°C–40 °C and 1.5–3.0, respectively. However, there are certain exceptionst, such as the optimal temperature for thermophilic *S. thermosulfidooxidans* being

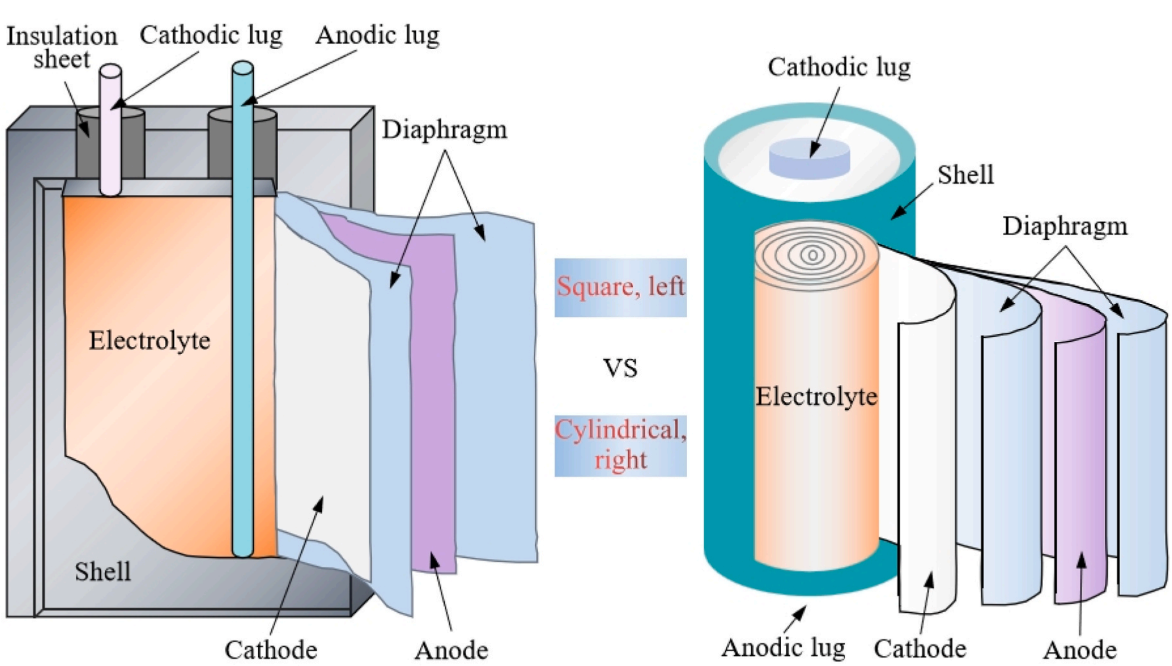

Fig. 1. Two typical lithium-ion batteries.

50 °C–55 °C and the appropriate pH for the fungus *A. niger* is 3.0–8.0. The bacteria and archaea among them perform Fe^{2+} oxidation (Eq. (1)), sulfur oxidation (Eq. (2)), carbon fixation (Eq. (3)), or nitrogen fixation (Eq. (4)), whereas *A. niger* performs bioleaching function mainly by the secretion of organic biological acids. Although such non-pathogenic microorganisms do not endanger human health, precautions should be taken to avoid contamination of the environment and foreign microorganisms when pure culture is studied in the laboratory. Therefore, laboratory equipments should be sterilized and dried beforehand, and experiments should be conducted under sterile and isolated conditions, such as clean benches.

$$4Fe^{2+} + 4H^{+} + O_2 \xrightarrow{Microbe} 4Fe^{3+} + 2H_2O \tag{1}$$

$$S^0 + O_2 + 2H_2O \xrightarrow{Microbe} SO_4^{2-} + 4H^{+} \tag{2}$$

$$6CO_2 + 12NADPH_2 + 18ATP \xrightarrow{Microbe} C_6H_{12}O_6 + 12NADP + 18ADP + 18Pi \tag{3}$$

$$N_2(+ 8H^{+} + 8e^{-} + 16ATP) \xrightarrow{Microbe} 2NH_3(+ H_2 + 16ADP + 16Pi) \xrightarrow{Microbe} Amino\ acid\ or\ NO_3^{-} \tag{4}$$

3.2. Bioleaching mechanism

After decades of development, bioleaching technology has been used to recover metals from feedstocks such as metal ores, printed circuit boards, fly ash, waste catalysts, and river sludge (Vera et al., 2013; Seidel et al., 2001; Gu et al., 2017). The bioleaching mechanism has been constantly evolving since Silverman and Ehrlich (1964) proposed the direct/indirect mechanism. Currently, most scholars prefer contact/non-contact mechanisms (Sand and Gehrke, 1999; Rohwerder and Sand, 2007). Acid leaching and complexation mechanisms have been considered as part of the bioleaching mechanism in recent years, especially for the dissolution of gold and gangue minerals (Li et al., 2021). In general, based on the metabolic activities of microorganisms, bioleaching allows metal elements to enter the solution in an ionic form, as shown in Fig. 4 (Panda et al., 2015). It has been shown that microorganisms are more efficient in their contact mechanism when they connect to the material surface using extracellular polymers (EPS). However, genes associated with EPS secretion are missing in some microorganisms, such as *Ferroplasma*. spp., whose adhesion may be mediated by EPS produced by other microorganisms (Rohwerder and Sand, 2007).

The bioleaching of spent LIBs has received attention due to both resource constraints and environmental concerns. The autotrophic *A. ferrooxidans* was used to leach $LiCoO_2$ cathode material, and the presence of microorganisms increased the leaching rate of Co by 24% (Mishra et al., 2008a). However, because low-valence metal compounds are not present in LIBs and Fe^{3+} oxidation does not directly contribute to metal release, Fe^{3+}-mediated chemical oxidation cannot explain LIBs bioleaching. Several studies indicated that lithium has a high leaching rates (>95%) but not for nickel, cobalt, or manganese (<70%), indicating significant differences in the bioleaching mechanism between lithium and the other three metals. It has been demonstrated that the Li–O bond in the battery is weaker than the Co–O bond, and therefore lithium is more readily leached (Bahaloo-Horeh et al., 2018; Cai et al., 2021).

The efficiency and mechanism of metal element release from LIBs at 1% concentration were explored for the first time with autotrophic *A. ferrooxidans*, *A. thiooxidans*, and *L. ferriphilum*. The non-contact bioleaching mechanism was found to play a dominant role in the leaching process. It was concluded that lithium release is achieved by the acid dissolution of H^{+} produced by microbial metabolism (Eq. (5)), whereas the three high-valent nickel, cobalt, and manganese require a combination of reduction and acid dissolution (Eq. (6)). Further studies concluded that Fe^{2+} is the key in direct reduction reactions (Eq. (7)), while Fe^{3+} is not involved in this process, implying that LIBs and ore bioleaching undergo different biochemical pathways (Baniasadi et al., 2019; Zeng et al., 2014; Wu et al., 2019). However, they share similarities, i.e., microbially mediated generation of the reducing agent Fe^{2+} (Eq. (8)) and acid production via sulfur biooxidation (Eq. (2)). Furthermore, the Fe^{3+} generated by biooxidation also decomposes pyrite through the chemical oxidation pathway (Eq. (9)). Under constant H^{+} supplementation, the bioleaching rates of Li, Ni, Co, and Mn all exceeded 95% (Xin et al., 2009, 2016).

Acid leaching and complexation (Eq. (10)) mediated by heterotrophic *A. niger* were used to dispose Ni–Co–Mn ternary LIBs. Citric acid has a greater leaching effect compared to other organic acids (gluconic, oxalic and malic acids) (Bahaloo-Horeh et al., 2018). However, heterotrophic microorganisms are adapted to low solid-liquid ratios, and the necessary organic nutrients are expensive, so research on LIBs bioleaching focuses on autotrophic microorganisms. In summary, LIBs bioleaching is driven by indirect mechanisms, and the biochemical pathway when pyrite is used as a nutrient is shown in Fig. 5 (Xin et al., 2009). It is noteworthy that organic acids produced by certain fungi can modify the valence state of metals and hence accelerate their dissolution

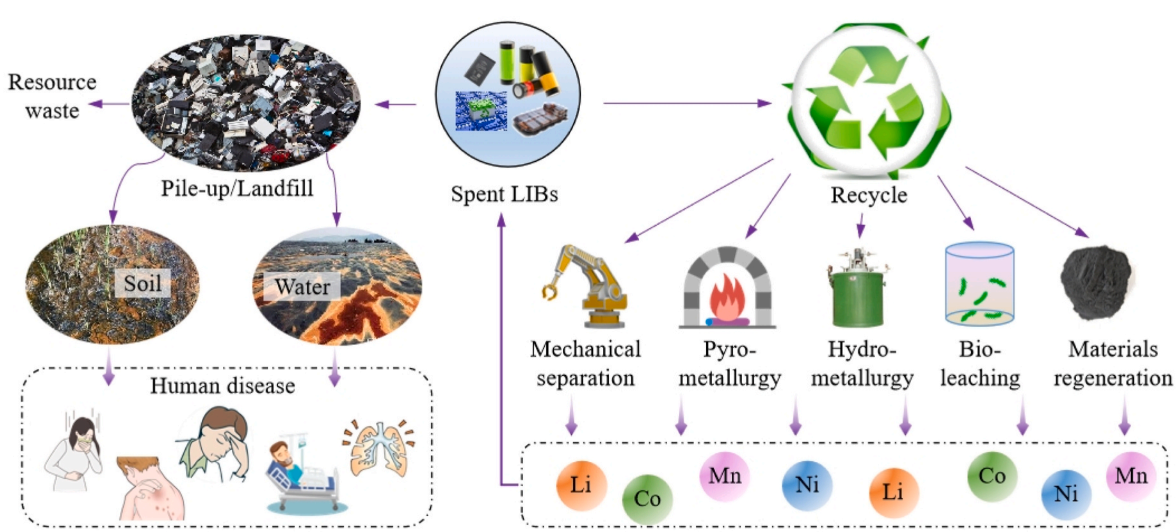

Fig. 2. Disposal status of spent lithium-ion batteries.

(for example, oxalic acid can reduce Fe^{3+} to Fe^{2+}) (Arslan and Bayat, 2009). Up to now, there have been no published reports showing the significance of organic acid reduction in the LIBs bioleaching.

$$Li_2O + H^+ \rightarrow 2Li^+ + H_2O \tag{5}$$

$$MO + 2H^+ \rightarrow M^{2+} + H_2O \tag{6}$$

$$M^{4+}/M^{3+} + Fe^{2+} \rightarrow M^{2+}(MO) + Fe^{3+} \tag{7}$$

$$FeS_2 + O_2 + 4H^+ \xrightarrow{Microbe} Fe^{2+} + 2S^0 + 2H_2O \tag{8}$$

$$FeS_2 + Fe^{3+} + 8H_2O \rightarrow 2Fe^{2+} + 2SO_4^{2-} + 16H^+ \tag{9}$$

$$MO + 2Ac^- + 2H^+ \rightarrow M(Ac)_2 + H_2O \tag{10}$$

where M denotes Li, Ni, Co or Mn elements and Ac^- denotes organic acid anion.

3.3. Stress response of microorganisms under metal toxicity

LIBs bioleaching systems trigger a series of stress responses in microorganisms, as illustrated in Fig. 6. Under normal conditions, cells continuously produce reactive oxygen species (ROS), causing malondialdehyde (MDA) to accumulate and be regarded as a biomarker of oxidative stress and peroxidation (Zheng et al., 2018b). Low levels of ROS are essential for cells to perform their functions because they regulate numerous crucial cellular pathways. During normal growth, the free radical scavenging system of microorganisms promptly scavenges excess ROS to achieve a dynamic balance between ROS production and scavenging (Wu et al., 2020; Wang et al., 2015). According to studies, the expression of genes related to energy metabolism in *A. ferrooxidans* is down-regulated under acid stress, and ATP synthase activity is significantly inhibited, leading to a drop in intracellular ATP level (Liu et al., 2022; Chao et al., 2008). High concentrations of nickel ions limit the function of the endogenous free radical scavenging system (Chen et al., 2023). Co^{2+} reacts with oxygen or hydrogen peroxide to produce excess ROS through a Fenton-type reaction. It causes intracellular redox imbalance and permanent damage to intracellular nucleic acids, proteins, lipids, carbohydrates, cell membranes, and other biomolecular components, potentially leading to microbial apoptosis (Liu et al., 2022; Gülçin, 2012).

Another study found a similar result: Li^+ and Co^{2+} led to an increase in solution osmotic pressure, which induced a large production of ROS in the colony, thus inhibiting microbial activity. Li^+ inhibited H_2O_2 scavenging ability and Co^{2+} inhibited O_2 scavenging ability of microorganisms. At the same time, they weakened the antioxidant enzymes of the colony, reducing the enzymatic activity of superoxide dismutase (SOD) and catalase (CAT). These processes lead to the accumulation of a large amount of ROS in the cells and trigger oxidative stress reactions, thereby inhibiting microbial activity and eventually leading to "slurry concentration limitation" in the $LiCoO_2$ bioleaching process (Wu et al., 2020). In addition, some scholars focused on the hydrophobicity of microorganisms and found that the planktonic cells in medium had a hydrophobicity of 29.5%, whereas sessile cells had a hydrophobicity of 70.2%. Furthermore, under the stress condition of 5.0% Li^+/Co^{2+}, the hydrophobicity of sessile cells decreased to 51.4% (Liu et al., 2022).

3.4. Process intensification

Numerous studies have shown that bioleaching is an effective way to recover valuable metals from spent LIBs, but it has not yet been successfully applied commercially, mainly for the following reasons: (1) Low microbial tolerance to metal ions. Much work has revealed that the microbial tolerance concentration threshold for LIBs leaching is 4%. That limits the processing efficiency of battery samples while also increasing production load and a variety of supporting expenditures. In industrial production, sample concentrations are generally required to be above 10% (Wang et al., 2018; Rohwerder et al., 2003). (2) Low leaching rate. Metal leaching rates of more than 70% in LIBs bioleaching are challenging to accomplish, especially for high-valence nickel, cobalt, and manganese (Bahaloo-Horeh et al., 2018; Cai et al., 2021). (3) Long leaching period. Numerous experimental results have shown that the bioleaching process requires 10–15 days to reach a stable leaching rate, which is a significant time investment (Boxall et al., 2018a; Jegan Roy et al., 2021). In order to improve the bioleaching efficiency, the influencing factors need to be strictly controlled and optimized, and the relevant contents have been well summarized (Sethurajan and Gaydardzhiev, 2021). Researchers have also proposed a series of enhancement methods to address the above issues in order to promote the industrial application of the technology.

3.4.1. Microbial activation

As mentioned above, metal ions dissolved from LIBs can poison microorganisms and weaken their catalytic action. Microbial tolerance can be improved by gradient domestication (Wang et al., 2018). Bahaloo-Horeh et al. (2018) found that domestication reduces the time required for microbial reproduction to enter the logarithmic phase and that the adaptation of *A. niger* to heavy metals increased the yield of organic acids and the metal leaching efficiency. Due to the absence of substances required for microbial growth in the battery, the addition of a nutritional substrate to the system is necessary. Sulfur provides energy and leaching agent (H^+) for the system through bio-oxidation (Eq. (2)), which can improve the efficiency of lithium leaching. However, it has no significant reductive promoting effect on the dissolution of higher-valence nickel, cobalt, and manganese compounds. Fe^{2+} is both an electron donor for microorganisms (Eq. (1)) and a reducing agent for

Table 1
Characteristics and application of recycling technologies for spent lithium-ion batteries.

Technology	Advantage	Disadvantage	Product	Application/Remarks
Crushing-sorting	Simple operation, big processing capacity	High energy consumption, miscellaneous products	Metal, mixture	Pretreatment process
Pyrometallurgy	Simple and mature, no sorting, strong adaptability	High temperature, high energy consumption, second pollution, high equipment requirements, lithium loss, need further refining	Alloy	Inmetco (America), Akkuser (Finland), Accurec/GRS Batterien (Germany), SNAM (Italy), Sony/Sumitomo/Mitsubishi (Japan)
Hydrometallurgy	High recovery rate, mild condition, diverse products	High reagent cost, long process, equipment corrosion, safety risk	Metal salt solution	GEM/Brunp (China), AEAT (England), Recupyl (France), Fortum (Finland)
Bioleaching	Low cost, low energy consumption, low pollution, mild condition	Long period, strict condition, low recovery rate	Metal salt solution	Not yet commercialized
Pyrometallurgy + Hydrometallurgy	No need for complex pretreatment processes, High recovery rate	Long process	Metal salt solution	Umicore (Belgium), Retriev (America), IME (Germany), Batrec (Switzerland)
Materials regeneration	Short process, full recovery	High raw materials requirements	Cathode material	Onto (America)

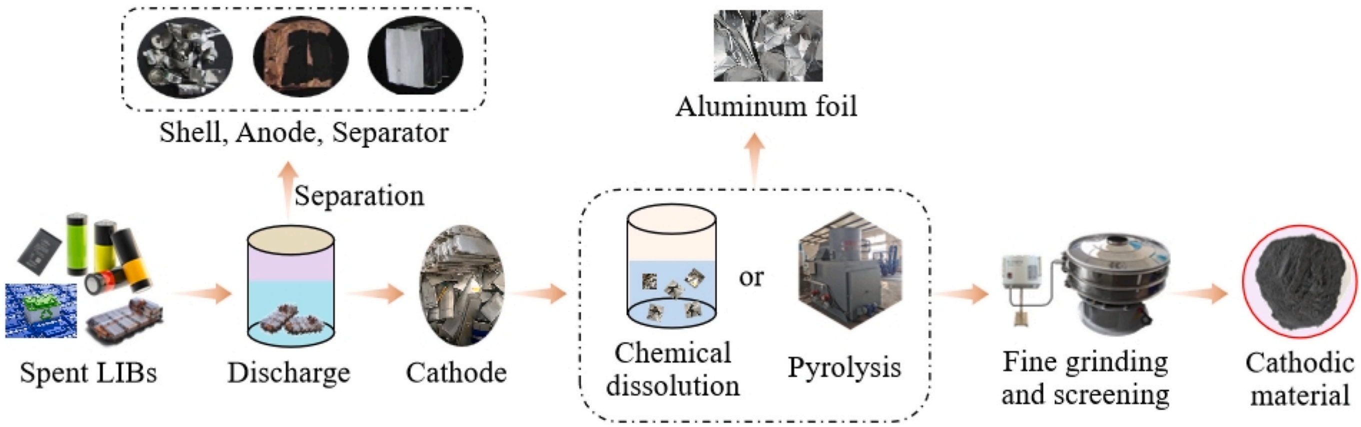

Fig. 3. Conventional pretreatment process for spent lithium-ion batteries.

Table 2
Common microorganisms for lithium-ion batteries bioleaching.

Category	Designation	Property			Function	Result	Reference
		Temperature	pH	Nutrition type			
Bacteria	*A. ferrooxidans*	30–35 °C	1.5–2.5	Autotrophic	Fe^{2+}/sulfur oxidation, Carbon/nitrogen fixation	100%Li, 88%Co, 20%Mn	Naseri et al. (2019a)
	Acidithiobacillus thiooxidans (A. thiooxidans)	28–30 °C	2.0–3.0	Autotrophic	Sulfur oxidation, Carbon fixation	90%Li, 12%Ni/Co, 20%Mn	Xin et al. (2016)
	Leptospirillum ferriphilum (L. ferriphilum)	30–40 °C	1.5–2.0	Autotrophic	Fe^{2+} oxidation, Carbon fixation	31%Li, 23%Ni/Co, 42%Mn	
	Acidithiobacillus caldus (A. caldus)	40–45 °C	2.0–2.5	Autotrophic	Sulfur oxidation, sulfide reduction	45%Li, 28%Co	Xin et al. (2016)
	S. thermosulfidooxidans	50–55 °C	1.9–2.4	Mixotrophic	Fe^{2+}/sulfur oxidation	59%Li, 35%Co	
Archaea	*Ferroplasma. spp*	35–36 °C	1.7	Heterotrophic	Fe^{2+} oxidation	73%Li, 98%Ni, 79% Co,	Ghassa et al. (2020)
Fungi	*A. niger*	25–35 °C	3.0–8.0	Heterotrophic	Organic acid secretion	95%Li, 38%Ni, 45% Co, 70%Mn	Bahaloo Horeh et al. (2016)

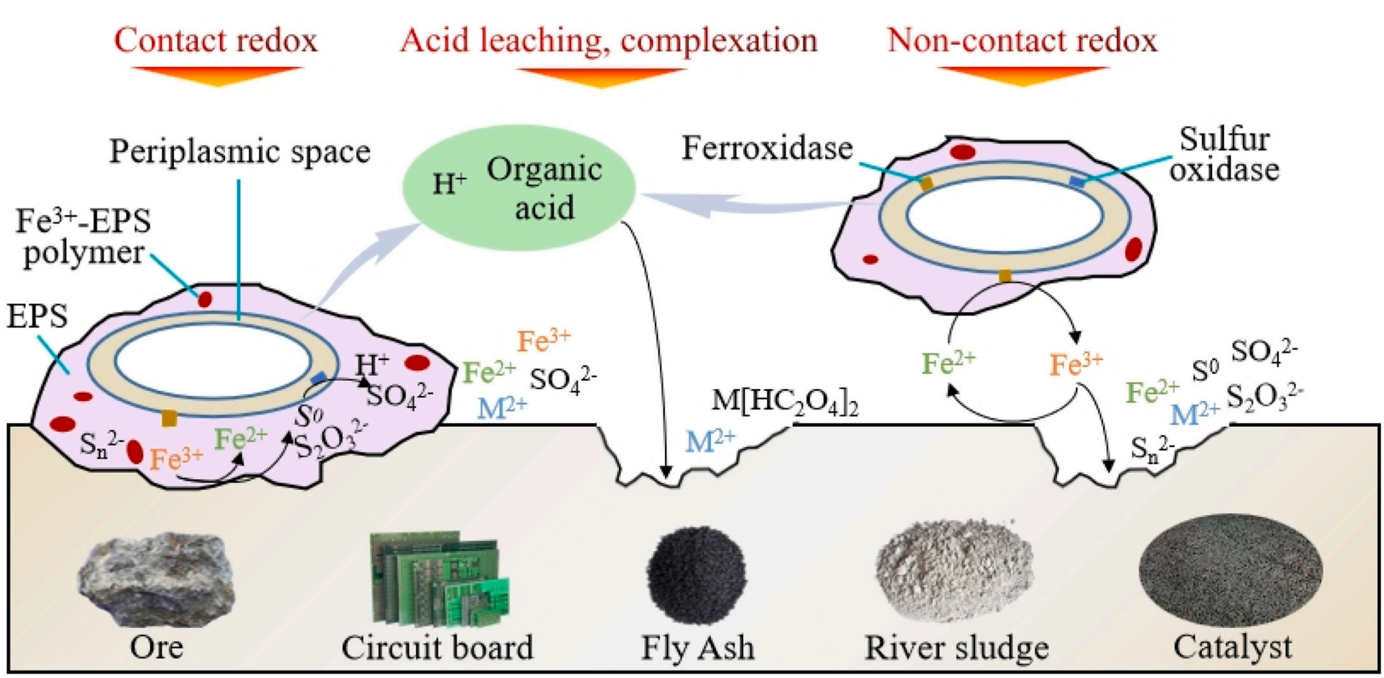

Fig. 4. Schematic diagram of the bioleaching mechanism for metal solid waste. Improved drawing based on (Jia et al., 2019).

the higher-valence metal oxides in the battery (Eq. (7)). Adding Fe^{2+} to the leaching system in the form of $FeSO_4$ can promote the leaching rates of nickel and cobalt to 99.7% and 99.9%, respectively, but this operation does not provide the leaching agent (H^+) (Ghassa et al., 2020). The provision of Fe^{2+} as pig iron (Fe) causes additional acid depletion and microbial apoptosis (Liao et al., 2022). Therefore, pyrite (FeS_2) as the

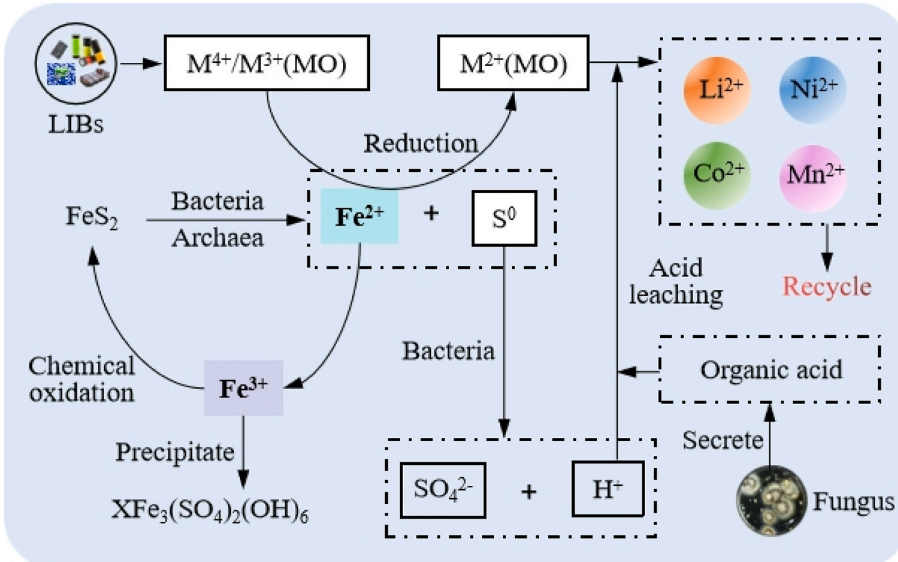

Fig. 5. Schematic diagram of the bioleaching mechanism of lithium-ion batteries.

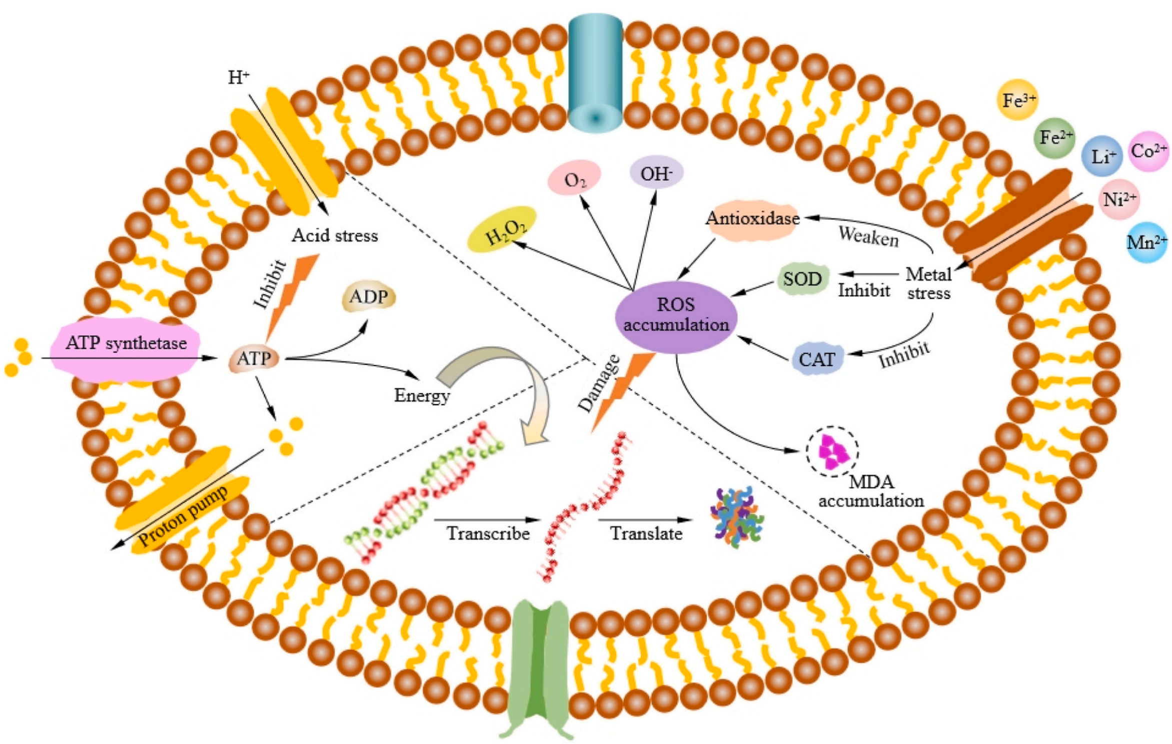

Fig. 6. The stress response of microorganisms in the bioleaching system of lithium-ion batteries (Liu et al., 2022; Chen et al., 2023; Liao et al., 2023).

nutritional substrate provides both H^+ and Fe^{2+} (Eq. (9)) and facilitates synergistic and effective polymetal bioleaching. In particular, the combined substrate of sulfur and pyrite can provide the system with a sufficient amount of H^+ and a strong change in redox potential (Xin et al., 2009). Some scholars boosted the dose of $FeSO_4$ in the 9 K medium by a factor of three, resulting in a high metal bioleaching (Do et al., 2022).

Mixed communities of autotrophic, heterotrophic, or mixed nutrient strains form a reciprocal relationship and outperform pure cultures in terms of leaching performance (Wu et al., 2020; Chaffron et al., 2010). Heterotrophic microorganisms can utilize low-molecular-weight organic matter as a metabolic substrate, dissipating their toxicity to autotrophic microorganisms. In this way, the microbial community can endure more severe bioleaching environments, allowing for higher slurry concentration (Sethurajan and Gaydardzhiev, 2021; Cai et al., 2021). Liao et al. (2022) inoculated a mixed culture of *A. caldus* and *S. thermosulfidooxidans* to increase metal leaching rates by 6%–20%. The same phenomenon occurred in the bioleaching recovery of waste circuit boards, where a blend of *A. ferrooxidans* and *A. thiooxidans* synergistically increased the leaching rates of copper, lead, and zinc by more than 20% compared to a pure culture (Wang et al., 2009). When LIBs are recovered in mixed cultures of autotrophic and heterotrophic microorganisms, a small amount (0.2 g/L) of yeast extract might prevent a decrease in microbial activity due to Fe^{2+} deficit (Wu et al., 2020).

To counteract peroxidative toxicity, there are ROS scavenging systems present in some microorganisms, such as *Leptospirillum ferrooxidans* (*L. ferrooxidans*), to enhance their stress resistance (Norambuena et al., 2012). Some ROS-scavenging enzymes, such as glutathione peroxidase (GSH-Px), superoxide dismutase (SOD), and catalase (CAT), are found in acidophilic microorganisms (Maaty et al., 2009). Exogenous antioxidants can thus be employed to mitigate the impact of oxidative stress on microorganisms. For example, spermine, glutathione, and cobalamin have been utilized to regulate intracellular ROS concentrations, chelate metal ions, and mitigate oxidative stress and physiological damage in microorganisms. Ultimately, a target leaching rate of >96% for both lithium and cobalt was achieved at a concentration of 5%, which was 10% higher than the control group. It was found that the biofilm state was poor under 5.0% Li^+/Co^{2+} stress, with very few live cells, and the addition of spermine resulted in thicker biofilms and more viable cells (Liu et al., 2020, 2022; Ferrer et al., 2016). Gallic acid was added to the LIBs bioleaching system to reduce metal toxicity, resulting in leaching rates of more than 98% for lithium and cobalt (Liao et al., 2023).

3.4.2. Process optimization

Process optimization is an essential and valuable strategy for improving the LIBs bioleaching efficiency. Two-step or multi-stage sequential bioleaching procedures have been proposed. The two-step method completes the microbial culture in the first stage and the metal leaching in the second. LIBs powder is added when the microorganisms have reached their maximal growth period (logarithmic growth period) to maximize microbial activity (Heydarian et al., 2018; Naseri et al., 2019b; Moazzam et al., 2021). A study of fly ash bioleaching found that one-step bioleaching is more effective for low concentrations (10–20 g/L), whereas two-step bioleaching is more effective for high concentrations (40–50 g/L) (Yang et al., 2008). Another study showed that a better leaching effect was achieved with cell-free media after separating microorganisms in the logarithmic growth phase from the system. In this way, both *A. niger* and *A. thiooxidans* resulted in an increase in lithium leaching rate of close to 10% (Biswal et al., 2018; Bahaloo-Horeh and Mousavi, 2017). The microorganisms are adapted to higher slurry concentrations by replacing fresh microbial solutions in a multi-stage continuous bioleaching process. This operation mitigates the acid depletion and metal ion toxicity of the waste solution, as well as the effects of passivation (Eq. (11)). In result, metal leaching rates increased by 10%–60%, particularly for lithium and manganese (Boxall et al., 2018b; Roy et al., 2021a, 2021b).

$$X^+ + 3Fe^{3+} + 2SO_4^{2-} + 6H_2O \rightarrow XFe_3(SO_4)_2(OH)_6\downarrow + 6H^+ \tag{11}$$

where X^+ denotes K^+, NH_4^+, or H_3O^+ ions.

Fine grinding is an effective way to facilitate the leaching process and can be explained by an increase in specific surface area (Bafghi et al., 2013). Furthermore, the mechanical activation effect generated by the high-energy ball milling process changes the microstructure and surface properties of the materials, e.g., by breaking chemical bonds, boosting leaching efficiency (Farhang et al., 2010; Yu et al., 2023). However, aggregation can affect leaching performance if the ball milling time is long (Zhao et al., 2021). Mechanical activation of cathode materials using a planetary grinding machine resulted in approximately 100% metal leaching rate under the action of L-ascorbic acid (Guo et al., 2018).

As described in the "Bioleaching Mechanisms" section, nickel, cobalt, and manganese are usually present in higher valence states in batteries, and their reduction to easily leachable lower valence compounds is necessary to achieve higher bioleaching efficiency. An electrochemical cathodic reduction method was proposed that utilizes electrons supplied by the cathode to achieve organic acid leaching of $LiCoO_2$ cathode materials, thus avoiding the usage of chemical reductants. Ultimately, both lithium and cobalt bioleaching rates exceeded 90% (Meng et al., 2018; Zhou et al., 2021). Based on the ion and electron transfer mechanisms, an electrokinetic approach combining bioleaching and selective adsorption of modified granular activated carbon was proposed, as shown in Fig. 7. In the bioleaching zone, bio-oxidation of pyrite and sulfur supplies the necessary reducing agent (Fe^{2+}) and additional leaching agent (H^+). The buffer zone avoids interfacial blockage and facilitates metal ion mobility within the electrolyzer. Co^{2+} is eventually trapped and chemically immobilized on the surface of the granular activated carbon, whereas the bulk of Li^+ and Mn^{2+} entering the cathode chamber. The maximum recoveries of 93.64%, 91.45%, and 87.92% were achieved for lithium, cobalt, and manganese, respectively (Huang et al., 2019). Similarly, Huang et al. (2013) proposed a microbial fuel cell method that avoided the consumption of large amounts of reducing agents and energy. The reducing environment in the cathode allowed the reduction of Co(III) to soluble Co(II), with a leaching rate that was 3.4 times that the sum of the conventional chemical and the acid-free control processes.

In addition to process improvements, optimization or control of parameters is equally important (Li et al., 2013a). Through single factor experiments, optimizing pH, temperature, energy substrate type, and slurry concentration increased the leaching rates of lithium and cobalt from approximately 60%, 30%–89%, 72%, respectively (Niu et al., 2014). Tanong et al. (2016) tested the leaching performance of inorganic acids, organic acids, chelating agents, and alkalis and found that sulfuric acid was the most effective solution for dissolving metals from spent LIBs. They further optimized the optimum leaching conditions using a three-layer Box-Behnken design. Algorithms have been used by some scholars to optimise process parameters. The support vector regression (SVR) modeling was used to maximize biomass generation to 25 g/L using *A. niger* as the inoculating microorganism (Ruhatiya et al., 2021). In the future, other algorithms based on genetic programming, deep learning neural networks, probabilistic and integrative learning may be used to further optimise the LIBs bioleaching process.

3.4.3. External catalysts

To improve the bioleaching efficiency, many catalysts are added to the leaching system to enhance microbial activity, elemental conversion, and electron transfer efficiency. The reduction of Fe^{2+} is critical in the bioleaching of LIBs, but adding $FeSO_4 \cdot 7H_2O$ continuously is not a good way to assure adequate Fe^{2+}. That would increase the workload, raise reagent costs, and cause passivation. This is because Fe^{2+} will be oxidized to Fe^{3+} under microbial catalysis, which not only weakens the reduction effect of Fe^{2+} on nickel, cobalt, and manganese high-valent oxides but also promote the generation of passivates like jarosite (Eq. (11)). Starch was used as a reducing agent and sulfuric acid as a leaching agent to recover spent LIBs, and the leaching rates of lithium, cobalt, nickel, and manganese reaching 98.55%, 96.73%, 97.6%, and 91.92%, respectively, but the required sulfuric acid concentration remained high (Lai et al., 2019). A study that directly used glucose as a reducing agent promoted the leaching efficiency of LIBs using phosphoric acid as a leaching agent (Meng et al., 2017). Zhang et al. (2018) found that catechins and epicatechins contained in grape seeds could be used as efficient reducing agents in the leaching process, significantly improving metal leaching efficiency when malic acid was used as a leaching agent. In one of our previous studies, hydrolysis of rice husk produced reducing sugars that promoted microbial colonization, improved community structure, and acted as a reducing agent, increasing the pyrite bioleaching rate by around 20% (Yang et al., 2021). Based on the above studies, redutive sugars in waste crops might be considered for inclusion in LIBs bioleaching systems to achieve synergistic multi-metal recovery and so meet the goal of “waste control by waste”.

Extracellular polymers (EPS) are natural high-molecular-weight polymers secreted by microorganisms in response to external stimuli and promote cell growth in unfavorable situations (Wang et al., 2020). As shown in Fig. 8, EPS has the ability to absorb metal elements and reduce their toxicity to microorganisms. In this way, it promotes cell growth under unfavorable conditions, resulting in the production of

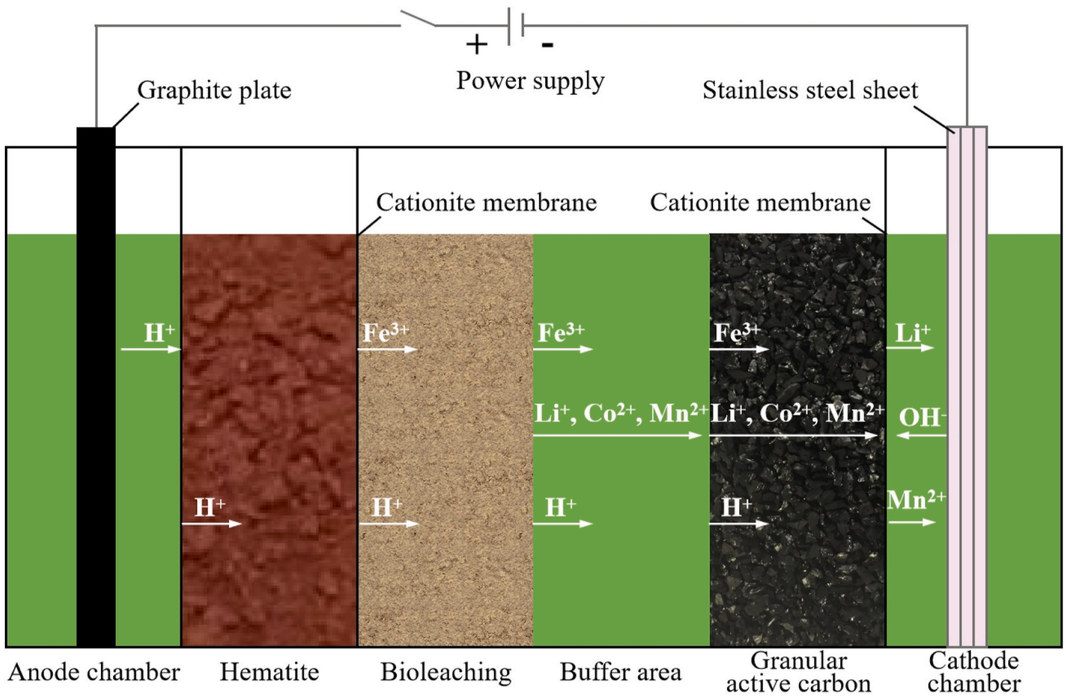

Fig. 7. Schematic diagram of the structure of the electrolytic biometallurgical plant and its ion migration (Huang et al., 2019).

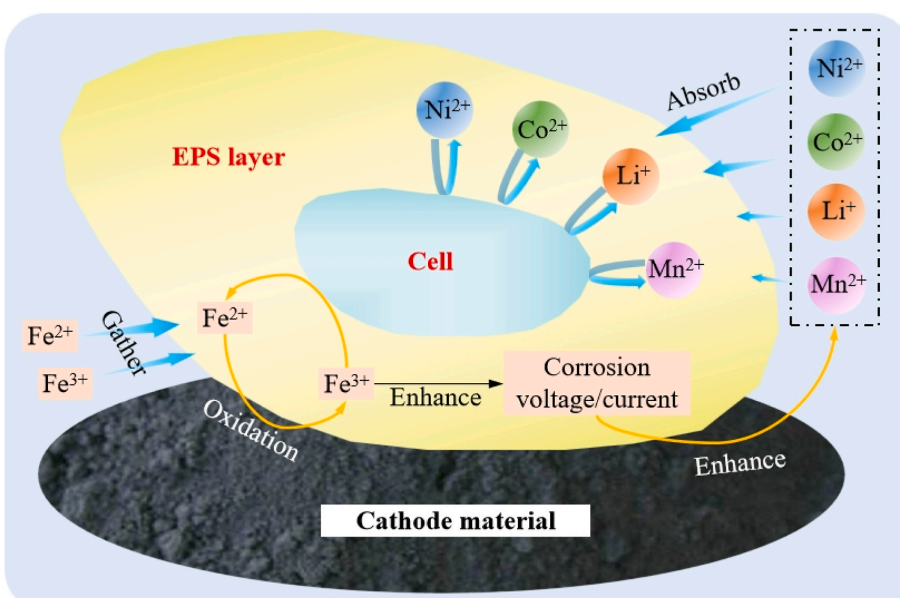

Fig. 8. Improvement mechanism of EPS on LIBs bioleaching.

more biological acid. Futhermore, Fe^{3+}/Fe^{2+} cycle can be enrichd through the non-contact mechanism. The high concentration of Fe^{2+} in the aggregator has a strong reducing effect on the cathode material, causing the synergistic release of high-valence nickel, cobalt, and manganese. EPS aggregation also greatly increases the corrosion voltage and thus generates higher corrosion currents (Wang et al., 2018; Kinzler et al., 2003). Studies have shown that the addition of EPS increases the bioleaching rates of nickel, cobalt, and manganese by 42%, 40%, and 44%, respectively (Wang et al., 2022).

In hydrometallurgical systems, the dissolution process can be accelerated by appropriate concentrations of metal ions. Different catalytic mechanisms have been proposed due to different material properties and crystalline forms, etc. It has been suggested that the enhancement of bioleaching efficiency by metal ions linked to enhanced electron transfer, an electrochemical phenomenon mediated by metal ions (Ballester et al., 1990; Vaughan, 1984). In pyrite-based bioleaching systems, the leaching rate of cobalt was greatly enhanced by the addition of Cu^{2+} and Ag^{+} (Zeng et al., 2012, 2013). The authors concluded that $LiCoO_2$ undergoes cation exchange reactions with metal ions on the sample surface, forming $CuCo_2O_4$ and $AgCoO_2$ intermediates (Eqs. (12) and (13)). Cobalt in the intermediate is oxidized further to Co^{2+} and then liberated (Eqs. (14) and (15)).

$$Cu^{2+} + 2LiCoO_2 \rightarrow CuCo_2O_4 + 2Li^{+} \tag{12}$$

$$Ag^{+} + LiCoO_2 \rightarrow AgCoO_2 + Li^{+} \tag{13}$$

$$CuCo_2O_4 + 6Fe^{3+} \rightarrow 6Fe^{2+} + Cu^{2+} + 2O_2 + 2Co^{2+} \quad (14)$$

$$AgCoO_2 + 3Fe^{3+} \rightarrow 3Fe^{2+} + Ag^{+} + 2O_2 + Co^{2+} \quad (15)$$

3.4.4. Omics technology and genetic engineering

To analyze the response of microbial endogenous factors to the external environment, omics such as genomics, proteomics, and metabolomics have been developed. A strain enriched out of sludge can reduce the bioleaching period of LIBs in half, and the macrogenome confirms that it contains a large number of genes encoding proteins that utilize organic compounds (Cai et al., 2021). Based on omics analysis, targeted modulation of microbial functional expression via genetic/metabolic engineering or external conditions is an essential strategy to enhance microbial tolerance. Tay et al. (2013) used proteomics to create two metabolically altered strains that resulted in a more than 50% increase in leachate secretion and a considerable increase in metal leaching rate. Yang et al. (2014) knocked out the gene coding of glucose oxidase for *Aspergillus carbonarius* (*A. carbonarius*), resulting in an increase in the amount of secreted citric, oxalic and malic acids. Therefore, studying the omics of LIBs bioleaching system can aid in artificially improving leaching efficiency.

To visualize the improvement of the LIBs bioleaching efficiency by enhancement measures, the relevant studies are listed in Table 3.

3.5. Leaching characteristics and interface phenomena

The bioleaching reaction of LIBs cathode material occurs at the solid-liquid interface, and metal dissolution is achieved through electron transfer. During this process, the elemental morphology, material surface, microbial state, and solution system are all changing dynamically. The $LiCoO_2$ X-ray diffraction (XRD) peak of the non-bioleaching group did not noticeably diminish, indicating that the effect of chemical leaching on the LIBs dissolution is negligible (Liao et al., 2022; Fang et al., 2022). When the acid dissolution is strong, all cobalt bivalent is released in the cathode materials, leaving only the acid-resistant cobalt trivalent with a narrow X-ray photoelectron spectroscopy peak (XPS), which ensures the integrity of the residue morphology. If there is also a reduction effect (Fe^{2+}), the cobalt trivalent is also released. At this time, the XPS peak of the cobalt compound is the narrowest, and the material particles are completely dissolved, resulting in corrosion pits. Further EDX analysis showed a significant decrease in cobalt content in the material's border, suggesting that bioleaching leads to structural destruction and boundary melting (Xin et al., 2009; Jegan Roy et al., 2021). When Cu^{2+} was added, the residue and solid product layers of larger particles disappeared, leaving only extremely sparse particles (jarosite) behind, indicating a better bioleaching effect (Zeng et al., 2012). Jarosite production was also confirmed by fourier transform infrared spectroscopy analysis (FTIR), which revealed round particles (Naseri et al., 2019a). Its synthesis may lead to passivation, preventing further leaching. However, some researchers looked at this issue from a different perspective. They suggested that the release of H^+ by the hydrolysis of Fe^{3+} into jarosite (Eq. (11)) or $Fe(OH)_3$ (Eq. (16)) could prevent a significant increase in the pH of the system and thus improve the dissolution of metal ions. At the same time, Fe^{3+} precipitation facilitates the subsequent removal of Fe^{3+} from leachate (Liao et al., 2022; Silva et al., 2020).

$$Fe^{3+} + 3H_2O \rightarrow Fe(OH)_3 + 3H^+ \quad (16)$$

Biofilms play an important role in resistance to metal ion stress and acid stress, and their formation is closely linked to bioleaching efficiency. Meanwhile, microorganisms are attached to the solid material by cell membranes. The number of sessile cells per unit volume may be 100 or 1000 more than that of planktonic cells, implying that biofilms can provide higher concentrations of metabolites on the ore surface, such as Fe^{3+} and H^+ (Liu et al., 2020). Under normal conditions, an adhering aggregated biofilm formed on the surface of the material, with a film thickness of 38.2 μm. However, the biofilm under 5.0% Li^+/Co^{2+} stress was incomplete, indicating that the structure and function of the biofilm have been greatly damaged, and the film thickness was only 12.1 μm. The addition of 1 mol/L spermine to the system resulted in a significant increase in live cells (Liu et al., 2020, 2022).

The changes in EPS composition during bioleaching were investigated using three-dimensional fluorescence spectroscopy (3DEEM), and it was found that the positive correlation coefficient between the concentration of EPS and manganese reached 0.99. According to the fluorescence regional integration (FRI), the total fluorescence area of EPS rose over time to 56550.9 au·nm^2. Transmission electron microscopy (TEM) analysis showed that microorganisms, EPS, and a large amount of bioleached material gathered together, and EPS has two important features: adsorption and bio-oxidation. This study confirms that pH adjustment and the addition of EPS can cause more severe structural damage to materials (Xin et al., 2016; Wang et al., 2022).

3.6. Process fitting and evaluation

Thermodynamics provides relevant information on the properties of the system during leaching equilibrium and the interaction between the

Table 3
Examples of enhancement studies for LIBs bioleaching.

LIBs material	Materials concentration, %	Temperature,° C	Time	Enhancement measures	Enhancement effect	References
$LiCoO_2$	5	42	72 h	Add 1 mmol/L spermine	Li and Co leaching rates increased by 9.8% and 11.8%, respectively	Liu et al. (2022)
NCM	4	30	60 min	Add EPS	Ni, Co, and Mn leaching rates increased by 7.1%, 5.7%, and 9.1%, respectively	Wang et al. (2018)
NCM	1	30	9 days	Sulfuric acid was added every 24 h to bring the pH to 1.0.	Ni and Co leaching rates increased by 58.7% and 52.5%, respectively	Xin et al. (2016)
$LiCoO_2$	1	35	6 days	Add 0.75 g/L Cu^{2+}	Co leaching rate increased by 56.8%	Zeng et al. (2012)
$LiCoO_2$	1	35	7 days	Add 0.02 g/L Ag^{2+}	Co leaching rate increased by 55.3%	Zeng et al. (2013)
NCM	10	30	3 days	Replacement of fresh bacterial solution every 24 h	Li, Ni, Co, and Mn leaching rates increased by 43%, 56%, 51%, and 61%, respectively	Jegan Roy et al. (2021)
NCM	1	30	30 days	Domestication	Ni, Co, and Mn leaching rates increased by 45%, 38%, and 64%, respectively	Bahaloo-Horeh et al., 2018
NCM	1	45	10 days	Add 24.25 g/L $FeSO_4{\cdot}H_2O$	Ni and Co leaching rates increased by 45% and 50%, respectively	Ghassa et al. (2020)
$LiCoO_2$	2	30	48 h	Mixed microorganism	Li and Co leaching rates increased by 7%–21% and 6%–13%, respectively	Liao et al. (2022)
$LiCoO_2$	2	30	48 h	Add 0.0075 mol/L gallic acid	Co leaching rate increased by 36.53%	Liao et al. (2023)
NCM	Solid-to-liquid radio: 40%		12 days	Bioelectrochemical leaching	The leaching rates of Li, Co, and Mn are 93.64%, 91.45%, and 87.92%	Huang et al. (2019)

system and the outside world when the state changes. The bioleaching rate of LIBs in solution is mainly limited by two factors: the redox potential and the pH. The E-pH diagram of the Li–Co–H_2O system shown in Fig. 9 shows that $LiCoO_2$ is converted to Co^{2+} and Li^+ when the redox potential is 0–1 V and the pH is 0–3 (bioleaching region, square frame). That is, the leached cobalt in solution exists as Co^{2+} rather than Co^{3+}, CoO_2, $Co(OH)_2$, or Co (Li et al., 2013a). Niu et al. (2014) calculated the bioleaching thermodynamics of cobalt and lithium from $LiCoO_2$ with sulfur and pyrite as mixed substrates. The Gibbs free energy of the bioleaching system is a huge negative values, 12.7 and 11.4 times higher than conventional chemical leaching with $FeSO_4$ and H_2O_2 as reactants, respectively. That is, bioleaching is more likely to occur spontaneously than chemical leaching.

The kinetic information of LIBs bioleaching process can be used to improve metal recovery rates. Based on the characteristics of the reaction process, the shrinkage unreacted core model was adopted, including boundary layer diffusion control (Eq. (17)), chemical reaction control (Eq. (18)), and product layer diffusion control (Eq. (19)) (Santos et al., 2010; Mishra et al., 2008b; Meshram et al., 2015; Azuma et al., 2019). Under specific conditions, these models can well describe the bioleaching behavior of lithium and cobalt, and the product layer diffusion model fits best (Niu et al., 2014; Sadeghabad et al., 2019). A study believes that the bioleaching behavior of manganese in LIBs fits better with the product layer diffusion model, whereas nickel and cobalt bioleaching fits better with the boundary layer diffusion model (Wang et al., 2022). Some scholars have updated the traditional model to account for fluctuations in microbial activity. The rate-limiting step for manganese was found to be the diffusion of iron ions through the liquid boundary layer surrounding the particles, whereas the rate-limiting step for lithium and cobalt was the diffusion of iron ions through the solid product layer (Naseri et al., 2019a).

$$x = k_1 t \tag{17}$$

$$1 - (1-x)^{1/3} = k_2 t \tag{18}$$

$$1 - 2\Big/3x - (1-x)^{2/3} = k_3 t \tag{19}$$

where k is the rate constant, x is the metal recovery rate, and t is the bioleaching time.

In order to clarify the LIBs bioleaching process, some scholars tested this process using electrochemical methods. Open circuit voltage, cyclic voltammetry, and Tafel modules were tested using powder microelectrodes. The presence of microorganisms was found to have a large effect on the corrosiveness of the cathode material, with a 0.09 μA/cm^2 increase in current density, suggesting that microorganisms increase the rate of cation exchange reactions. Further analysis revealed that lithium cobaltate began to decompose at lower potentials, with a corrosion potential of 0.420 V. When the passivation potential (0.776 V) is reached, the current density rapidly decreases, indicating the formation of a passivation film on the surface of $LiCoO_2$, and the final passivation is formed at 0.802 V (Li et al., 2013a; Deng et al., 2012). In one investigation, adding exogenous spermine to the bioleaching increased current density by 0.9 μA/cm^2, indicating accelerated biofilm formation and more severe pyrite corrosion (Liu et al., 2022).

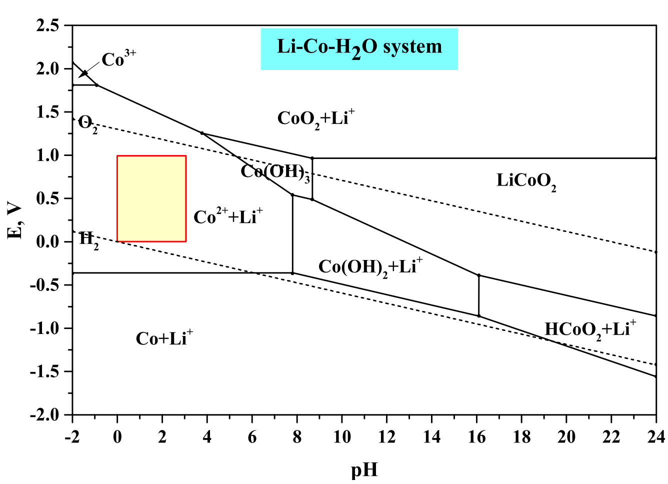

Fig. 9. E-pH diagram of Li–Co–H_2O system. All ion concentrations are 10^{-3} mol/L.

Technical evaluation should encompass not only production efficiency but also economic and environmental friendliness (Liang et al., 2021; Li et al., 2013b; Lv et al., 2018). Process simulation was used to find that the process with solvent extraction was more economically efficient than the other process (Granata et al., 2012). In the vast majority of the research literature, the claim that bioleaching methods are environmentally friendly is based only on process-centric assumptions. In fact, the environmental impact of various production processes can be quantified using the life cycle assessment method (LCA) (Heydarian et al., 2018). Villares et al. (2016) evaluated the environmental impact of pyrometallurgical and bioleaching methods for e-waste using the ex-ante LCA method. For $LiNi_{1/3}Co_{1/3}Mn_{1/3}O_2$ cathode material, one study quantified and compared the environmental impact of nine hydrometallurgical recycling processes on eighteen indicators, such as global warming when processing 1 Kg of cathode material using OpenLCA software. As shown in Fig. 10, recovery processes based on organic solvents (lactic or citric acid) show three times more greenhouse gases than processes using inorganic acids (HCl or H_2SO_4). This result is due to an increase in the use of chemical reagents (low solid-liquid ratio). The preferred processes for reducing greenhouse gas emissions and resource-related potential impacts include HCl, H_2SO_4/H_2O_2, and autotrophic bioleaching. Sensitivity analysis shows that the global warming value is reduced to 5.01 Kg (CO_2)/$Kg_{cathode}$ by the transition to renewable energy (Iturrondobeitia et al., 2022).

4. Conclusion and outlook

With the current explosive growth of new energy vehicles, LIBs will soon become the main e-waste source. Gradient utilisation, material recycling, and metal recovery are all sustainable development methods based on different considerations. Given the current global environmental problems, the environmental cost should be given more attention when considering ways to recycle LIBs. Therefore, the industrialization of the LIBs bioleaching is worthy of people's efforts.

The current development state of LIBs bioleaching is reviewed in this work, including microorganisms, bioleaching mechanisms, metal stress on microorganisms, process intensification, leaching characterization

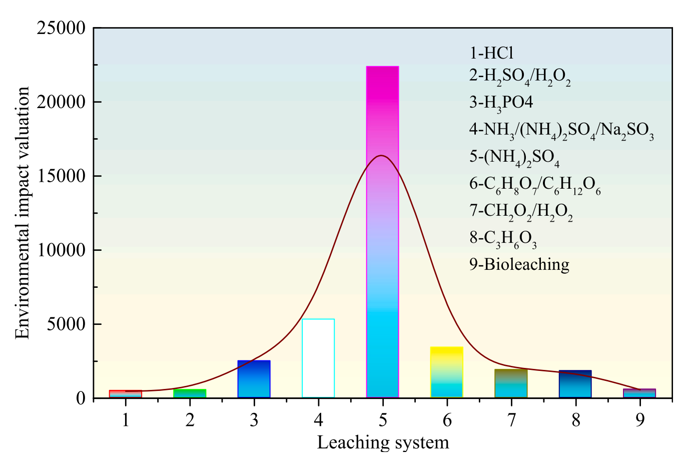

Fig. 10. Impact index of leaching system on the environment.

and interfacial phenomena, process fitting, and evaluation. The dissolution of metals in LIBs relies on the combined action of effective reducing and leaching agents. Metal ions exert toxic stress on microorganisms and interfere with biochemical processes like protein secretion. Microbial activation, process optimization, and external additives are the main measures for enhancing LIBs bioleaching. The massive secretion of EPS will increase the corrosion voltage/current, thus improving bioleaching efficiency. Compared with traditional hydrometallurgical processes, bioleaching shows greater advantages in environmental protection.

High microbial activity is the core of bioleaching and a prerequisite for good metal recovery. In the future, the development of omics and genetic engineering technologies may provide better support for LIBs bioleaching. They contribute to the discovery of new knowledge and characteristics of microorganisms regarding the interaction of genes, proteins, etc. with the environment. Microbial responses to metal stress are closely linked to responses at the genetic scale. Transcriptomics can explore the mechanisms of metabolic regulation at the genetic level and reveal relationships between responses at different scales. At the same time, omics aids in the advancement of scientific knowledge about the microbial physiology involved in bioleaching, including the metabolism of extreme microorganisms and the synergistic role of microorganisms in bioleaching systems. On this basis, strains can then be genetically or chemically manipulated to further improve their resistance to metal stress.

Due to the differences in microbial functions, microbial community regulation and metabolic control are also options to enhance the leaching activity of microorganisms. In mixed bioleaching systems for LIBs, the addition of reducing sugars, such as glucose, may have a surprising effect. This is because such reagents will not only convert high-valent insoluble metal compounds in the material to low-valent easily leachable compounds, but also provide nutrient substrates and regulate the community structure of heterotrophic microorganisms in the mixed community. What remains is consideration of the cost and operability of the production process.

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Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data availability

No data was used for the research described in the article.

References

Arslan, V., Bayat, O., 2009. Removal of Fe from kaolin by chemical leaching and bioleaching. Clay Clay Miner. 57, 787–794. https://doi.org/10.1346/CCMN.2009.05706011.

Azuma, D., Aikawa, T., Hiraga, Y., Watanabe, M., Smith Jr., R.L., 2019. Kinetic study of hydrothermal leaching of lithium cobalt oxide with citric acid. Kagaku Kogaku Ronbunshu 45, 147–157. https://doi.org/10.1252/kakoronbunshu.45.147.

Bafghi, M.S., Emami, A.H., Zakeri, A., 2013. Effect of specific surface area of a mechanically activated chalcopyrite on its rate of leaching in sulfuric Acid-Ferric sulfate media. Metall. Mater Trans. B-Proc. Metall Mater Proc. Sci. 44, 1166–1172. https://doi.org/10.1007/s11663-013-9890-0.

Bahaloo Horeh, N., Mousavi, S.M., Shojaosadati, S.A., 2016. Bioleaching of valuable metals from spent lithium-ion mobile phone batteries using *Aspergillus niger*. J. Power Sources 320, 257–266. https://doi.org/10.1016/j.jpowsour.2016.04.104.

Bahaloo-Horeh, N., Mousavi, S.M., 2017. Enhanced recovery of valuable metals from spent lithium-ion batteries through optimization of organic acids produced by *Aspergillus niger*. Waste Manage. (Tucson, Ariz.) 60, 666–679. https://doi.org/10.1016/j.wasman.2016.10.034.

Bahaloo-Horeh, N., Mousavi, S.M., Baniasadi, M., 2018. Use of adapted metal tolerant *Aspergillus niger* to enhance bioleaching efficiency of valuable metals from spent lithium-ion mobile phone batteries. J. Clean. Prod. 197, 1546–1557. https://doi.org/10.1016/j.jclepro.2018.06.299.

Ballester, A., González, F., Blázquez, M.L., Mier, J.L., 1990. The influence of various ions in the bioleaching of metal sulphides. Hydrometallurgy 23, 221–235. https://doi.org/10.1016/0304-386X(90)90006-N.

Baniasadi, M., Vakilchap, F., Bahaloo-Horeh, N., Mousavi, S.M., Farnaud, S., 2019. Advances in bioleaching as a sustainable method for metal recovery from e-waste: a review. J. Ind. Eng. Chem. 76, 75–90. https://doi.org/10.1016/j.jiec.2019.03.047.

Biswal, B.K., Jadhav, U.U., Madhaiyan, M., Ji, L., Yang, E., Cao, B., 2018. Biological leaching and chemical precipitation methods for recovery of co and li from spent Lithium-Ion batteries. Acs. Sustain. Chem. Eng. 6, 12343–12352. https://doi.org/10.1021/acssuschemeng.8b02810.

Boxall, N.J., Cheng, K.Y., Bruckard, W., Kaksonen, A.H., 2018a. Application of indirect non-contact bioleaching for extracting metals from waste lithium-ion batteries. J. Hazard Mater. 360, 504–511. https://doi.org/10.1016/j.jhazmat.2018.08.024.

Boxall, N.J., Cheng, K.Y., Bruckard, W., Kaksonen, A.H., 2018b. Application of indirect non-contact bioleaching for extracting metals from waste lithium-ion batteries. J. Hazard Mater. 360, 504–511. https://doi.org/10.1016/j.jhazmat.2018.08.024.

Cai, X., Tian, L., Chen, C., Huang, W., Yu, Y., Liu, C., Yang, B., Lu, X., Mao, Y., 2021. Phylogenetically divergent bacteria consortium from neutral activated sludge showed heightened potential on bioleaching spent lithium-ion batteries. Ecotoxicol. Environ. Saf. 223, 112592 https://doi.org/10.1016/j.ecoenv.2021.112592.

Chaffron, S., Rehrauer, H., Pernthaler, J., von Mering, C., 2010. A global network of coexisting microbes from environmental and whole-genome sequence data. Genome Res. 20, 947–959. https://doi.org/10.1101/gr.104521.109.

Chagnes, A., Pospiech, B., 2013. A brief review on hydrometallurgical technologies for recycling spent lithium-ion batteries. J. Chem. Technol. Biotechnol. 88, 1191–1199. https://doi.org/10.1002/jctb.4053.

Chandran, V., Ghosh, A., Patil, C.K., Mohanavel, V., Priya, A.K., Rahim, R., Madavan, R., Muthuraman, U., Karthick, A., 2021. Comprehensive review on recycling of spent lithium-ion batteries. Mater. Today: Proc. 47, 167–180. https://doi.org/10.1016/j.matpr.2021.03.744.

Chao, J., Wang, W., Xiao, S., Liu, X., 2008. Response of *Acidithiobacillus ferrooxidans* ATCC 23270 gene expression to acid stress. World J. Microbiol. Biotechnol. 24, 2103–2109. https://doi.org/10.1007/s11274-008-9715-5.

Chen, L., Tang, X., Zhang, Y., Li, L., Zeng, Z., Zhang, Y., 2011. Process for the recovery of cobalt oxalate from spent lithium-ion batteries. Hydrometallurgy 108, 80–86. https://doi.org/10.1016/j.hydromet.2011.02.010.

Chen, H., Gu, S., Guo, Y., Dai, X., Zeng, L., Wang, K., He, C., Dodbiba, G., Wei, Y., Fujita, T., 2021. Leaching of cathode materials from spent lithium-ion batteries by using a mixture of ascorbic acid and HNO_3. Hydrometallurgy 205, 105746. https://doi.org/10.1016/j.hydromet.2021.105746.

Chen, G., Shi, H., Ding, H., Zhang, X., Gu, T., Zhu, M., Tan, W., 2023. Multi-scale analysis of nickel ion tolerance mechanism for thermophilic *Sulfobacillus thermosulfidooxidans* in bioleaching. J. Hazard Mater. 443, 130245 https://doi.org/10.1016/j.jhazmat.2022.130245.

Deng, X., Zeng, G., Luo, S., Luo, X., Zou, J., 2012. Electrochemical behavior of bioleaching $LiCoO_2$ from spent lithium-ion batteries by *Thiobacillus ferrooxidans*. J. Cent. S. Univ. 43, 2500–2505 (In Chinese).

Do, M.P., Jegan Roy, J., Cao, B., Srinivasan, M., 2022. Green Closed-Loop cathode regeneration from spent NMC-Based Lithium-Ion batteries through bioleaching. Acs. Sustain. Chem. Eng. 10, 2634–2644. https://doi.org/10.1021/acssuschemeng.1c06885.

Fang, X., Sun, S., Liao, X., Li, S., Zhou, S., Gan, Q., Zeng, L., Guan, Z., 2022. Effect of diurnal temperature range on bioleaching of sulfide ore by an artificial microbial consortium. Sci. Total Environ. 806, 150234 https://doi.org/10.1016/j.scitotenv.2021.150234.

Farhang, M.R., Kamali, A.R., Nazarian-Samani, M., 2010. Effects of mechanical alloying on the characteristics of a nanocrystalline Ti–50at.%Al during hot pressing consolidation. Mater Sci. Eng. B-Solid State Mater Adv. Technol. 168, 136–141. https://doi.org/10.1016/j.mseb.2009.10.032.

Ferrer, A., Rivera, J., Zapata, C., Norambuena, J., Sandoval, Á., Chávez, R., Orellana, O., Levicán, G., 2016. Cobalamin protection against oxidative stress in the acidophilic iron-oxidizing bacterium *Leptospirillum* group II CF-1. Front. Microbiol. 7 https://doi.org/10.3389/fmicb.2016.00748.

Gao, G., He, X., Li, Y., Yang, J., Li, C., Xu, X., Xiao, H., Guan, J., 2017. Current status of recycling technology of spent automotive lithium-ion batteries. Environ. Eng. 35, 135–140. https://doi.org/10.13205/j.hjgc.201710028.

Georgi-Maschler, T., Friedrich, B., Weyhe, R., Heegn, H., Rutz, M., 2012. Development of a recycling process for Li-ion batteries. J. Power Sources 207, 173–182. https://doi.org/10.1016/j.jpowsour.2012.01.152.

Ghassa, S., Farzanegan, A., Gharabaghi, M., Abdollahi, H., 2020. Novel bioleaching of waste lithium ion batteries by mixed moderate thermophilic microorganisms, using iron scrap as energy source and reducing agent. Hydrometallurgy 197, 105465. https://doi.org/10.1016/j.hydromet.2020.105465.

Ghassa, S., Farzanegan, A., Gharabaghi, M., Abdollahi, H., 2021. Iron scrap, a sustainable reducing agent for waste lithium ions batteries leaching: an environmentally friendly method to treating waste with waste. Resour. Conserv. Recycl. 166, 105348 https://doi.org/10.1016/j.resconrec.2020.105348.

Golmohammadzadeh, R., Rashchi, F., Vahidi, E., 2017. Recovery of lithium and cobalt from spent lithium-ion batteries using organic acids: process optimization and kinetic aspects. Waste Manage. (Tucson, Ariz.) 64, 244–254. https://doi.org/10.1016/j.wasman.2017.03.037.

Granata, G., Moscardini, E., Pagnanelli, F., Trabucco, F., Toro, L., 2012. Product recovery from Li-ion battery wastes coming from an industrial pre-treatment plant: lab scale tests and process simulations. J. Power Sources 206, 393–401. https://doi.org/10.1016/j.jpowsour.2012.01.115.

Gu, W., Bai, J., Dong, B., Zhuang, X., Zhao, J., Zhang, C., Wang, J., Shih, K., 2017. Enhanced bioleaching efficiency of copper from waste printed circuit board driven by nitrogen-doped carbon nanotubes modified electrode. Chem. Eng. J. 324, 122–129. https://doi.org/10.1016/j.cej.2017.05.024.

Gülçin, İ., 2012. Antioxidant activity of food constituents: an overview. Arch. Toxicol. 86, 345–391. https://doi.org/10.1007/s00204-011-0774-2.

Guo, Y., Li, Y., Lou, X., Guan, J., Li, Y., Mai, X., Liu, H., Zhao, C.X., Wang, N., Yan, C., Gao, G., Yuan, H., Dai, J., Su, R., Guo, Z., 2018. Improved extraction of cobalt and lithium by reductive acid from spent lithium-ion batteries via mechanical activation process. J. Mater. Sci. 53, 13790–13800. https://doi.org/10.1007/s10853-018-2229-0.

Heydarian, A., Mousavi, S.M., Vakilchap, F., Baniasadi, M., 2018. Application of a mixed culture of adapted acidophilic bacteria in two-step bioleaching of spent lithium-ion laptop batteries. J. Power Sources 378, 19–30. https://doi.org/10.1016/j.jpowsour.2017.12.009.

Huang, L., Li, T., Liu, C., Quan, X., Chen, L., Wang, A., Chen, G., 2013. Synergetic interactions improve cobalt leaching from lithium cobalt oxide in microbial fuel cells. Bioresour. Technol. 128, 539–546. https://doi.org/10.1016/j.biortech.2012.11.011.

Huang, T., Liu, L., Zhang, S., 2019. Recovery of cobalt, lithium, and manganese from the cathode active materials of spent lithium-ion batteries in a bio-electro-hydrometallurgical process. Hydrometallurgy 188, 101–111. https://doi.org/10.1016/j.hydromet.2019.06.011.

Iturrondobeitia, M., Vallejo, C., Berroci, M., Akizu-Gardoki, O., Minguez, R., Lizundia, E., 2022. Environmental impact assessment of $LiNi_{1/3}Mn_{1/3}Co_{1/3}O_2$ Hydrometallurgical cathode recycling from spent Lithium-Ion batteries. Acs Sustain. Chem. Eng. 10, 9798–9810. https://doi.org/10.1021/acssuschemeng.2c01496.

Jegan Roy, J., Srinivasan, M., Cao, B., 2021. Bioleaching as an Eco-Friendly approach for metal recovery from spent NMC-Based Lithium-Ion batteries at a high pulp density. Acs Sustain. Chem. Eng. 9, 3060–3069. https://doi.org/10.1021/acssuschemeng.0c06573.

Jia, Y., Tan, Q., Sun, H., Zhang, Y., Gao, H., Ruan, R., 2019. Sulfide mineral dissolution microbes: community structure and function in industrial bioleaching heaps. Green Energy Environ. 4, 29–37. https://doi.org/10.1016/j.gee.2018.04.001.

Kinzler, K., Gehrke, T., Telegdi, J., Sand, W., 2003. Bioleaching—a result of interfacial processes caused by extracellular polymeric substances (EPS). Hydrometallurgy 71, 83–88. https://doi.org/10.1016/S0304-386X(03)00176-2.

Lai, Y., Yang, J., Zhang, G., Tang, Y., Jiang, L., Yang, S., Li, X., 2019. Optimization and kinetics of leaching valuable metals from cathode materials of spent ternary lithium ion batteries with starch as reducing agent. Chin. J. Nonferrous Metals 29, 153–160. https://doi.org/10.19476/j.ysxb.1004.0609.2019.01.18.

Lai, X., Huang, Y., Gu, H., Deng, C., Han, X., Feng, X., Zheng, Y., 2021. Turning waste into wealth: a systematic review on echelon utilization and material recycling of retired lithium-ion batteries. Energy Storage Mater. 40, 96–123. https://doi.org/10.1016/j.ensm.2021.05.010.

Lee, S., Lee, D., Lee, S., Sohn, H., 2022. Lithium recovery from NCM lithium-ion battery by carbonation roasting with graphite followed by water leaching. Resour. Recycl. 31, 26–33. https://doi.org/10.7844/kirr.2022.31.4.26.

Li, L., Ge, J., Wu, F., Chen, R., Chen, S., Wu, B., 2010. Recovery of cobalt and lithium from spent lithium ion batteries using organic citric acid as leachant. J. Hazard Mater. 176, 288–293. https://doi.org/10.1016/j.jhazmat.2009.11.026.

Li, L., Zeng, G., Luo, S., Deng, X., Xie, Q., 2013a. Influences of solution pH and redox potential on the bioleaching of $LiCoO_2$ from spent lithium-ion batteries. J. Korean Soc. Appl. Bi 56, 187–192. https://doi.org/10.1007/s13765-013-3016-x.

Li, L., Dunn, J.B., Zhang, X.X., Gaines, L., Chen, R.J., Wu, F., Amine, K., 2013b. Recovery of metals from spent lithium-ion batteries with organic acids as leaching reagents and environmental assessment. J. Power Sources 233, 180–189. https://doi.org/10.1016/j.jpowsour.2012.12.089.

Li, L., Zhang, X., Li, M., Chen, R., Wu, F., Amine, K., Lu, J., 2018. The recycling of spent Lithium-Ion batteries: a review of current processes and technologies. Electrochem. Energy Rev. 1, 461–482. https://doi.org/10.1007/s41918-018-0012-1.

Li, J., Yang, H., Tong, L., 2021. Microbial solubilization of gangue minerals and their influence on pyrite bio-oxidation. Geochemistry 81, 125790. https://doi.org/10.1016/j.chemer.2021.125790.

Li, J., Yang, H., Tong, L., Sand, W., 2022. Some aspects of industrial heap bioleaching technology: from basics to practice. Min. Proc. Ext. Met. Rev. 43, 510–528. https://doi.org/10.1080/08827508.2021.1893720.

Li, W., Chen, Y., Li, S., Wang, C., Li, Y., Zhao, T., Traversy, M., Chang, C., Jie, Y., He, J., Tang, C., Yang, S., 2023. Leaching kinetics of de-lithium residue from spent ternary lithium-ion battery cathodic materials with starch as reductant. T Nonferr. Metal. Soc. 33, 619–631. https://doi.org/10.1016/S1003-6326(22)66133-8.

Liang, Z., Cai, C., Peng, G., Hu, J., Hou, H., Liu, B., Liang, S., Xiao, K., Yuan, S., Yang, J., 2021. Hydrometallurgical recovery of spent lithium ion batteries: environmental strategies and sustainability evaluation. Acs Sustain. Chem. Eng. 9, 5750–5767. https://doi.org/10.1021/acssuschemeng.1c00942.

Liao, X., Ye, M., Liang, J., Guan, Z., Li, S., Deng, Y., Gan, Q., Liu, Z., Fang, X., Sun, S., 2022. Feasibility of reduced iron species for promoting Li and Co recovery from spent $LiCoO_2$ batteries using a mixed-culture bioleaching process. Sci. Total Environ. 830, 154577 https://doi.org/10.1016/j.scitotenv.2022.154577.

Liao, X., Ye, M., Liang, J., Jian, J., Li, S., Gan, Q., Liu, Z., Mo, Z., Huang, Y., Sun, S., 2023. Comprehensive insights into the gallic acid assisted bioleaching process for spent LIBs: relationships among bacterial functional genes, Co(III) reduction and metal dissolution behavior. J. Hazard Mater. 447, 130773 https://doi.org/10.1016/j.jhazmat.2023.130773.

Liu, X., Liu, H., Wu, W., Zhang, X., Gu, T., Zhu, M., Tan, W., 2020. Oxidative stress induced by metal ions in bioleaching of $LiCoO_2$ by an acidophilic microbial consortium. Front. Microbiol. 10 https://doi.org/10.3389/fmicb.2019.03058.

Liu, D., Shi, H., Chen, G., Zhang, X., Gu, T., Zhu, M., Tan, W., 2022. Strategies for anti-oxidative stress and anti-acid stress in bioleaching of $LiCoO_2$ using an acidophilic microbial consortium. Extremophiles 26. https://doi.org/10.1007/s00792-022-01270-3.

Lv, W., Wang, Z., Cao, H., Sun, Y., Zhang, Y., Sun, Z., 2018. A critical review and analysis on the recycling of spent Lithium-Ion batteries. Acs Sustain. Chem. Eng. 6, 1504–1521. https://doi.org/10.1021/acssuschemeng.7b03811.

Maaty, W.S., Wiedenheft, B., Tarlykov, P., Schaff, N., Heinemann, J., Robison-Cox, J., Valenzuela, J., Dougherty, A., Blum, P., Lawrence, C.M., Douglas, T., Young, M.J., Bothner, B., 2009. Something old, something new, something borrowed; How the thermoacidophilic archaeon *sulfolobus solfataricus* responds to oxidative stress. PLoS One 4, e6964. https://doi.org/10.1371/journal.pone.0006964.

Meng, Q., Zhang, Y., Dong, P., 2017. Use of glucose as reductant to recover Co from spent lithium ions batteries. Waste Manage. (Tucson, Ariz.) 64, 214–218. https://doi.org/10.1016/j.wasman.2017.03.017.

Meng, Q., Zhang, Y., Dong, P., 2018. Use of electrochemical cathode-reduction method for leaching of cobalt from spent lithium-ion batteries. J. Clean. Prod. 180, 64–70. https://doi.org/10.1016/j.jclepro.2018.01.101.

Meshram, P., Pandey, B.D., Mankhand, T.R., 2015. Recovery of valuable metals from cathodic active material of spent lithium ion batteries: leaching and kinetic aspects. Waste Manage. (Tucson, Ariz.) 45, 306–313. https://doi.org/10.1016/j.wasman.2015.05.027.

Mishra, D., Kim, D., Ralph, D.E., Ahn, J., Rhee, Y., 2008a. Bioleaching of metals from spent lithium ion secondary batteries using *Acidithiobacillus ferrooxidans*. Waste Manage. (Tucson, Ariz.) 28, 333–338. https://doi.org/10.1016/j.wasman.2007.01.010.

Mishra, D., Kim, D.J., Ralph, D.E., Ahn, J.G., Rhee, Y.H., 2008b. Bioleaching of spent hydro-processing catalyst using acidophilic bacteria and its kinetics aspect. J. Hazard Mater. 152, 1082–1091. https://doi.org/10.1016/j.jhazmat.2007.07.083.

Moazzam, P., Boroumand, Y., Rabiei, P., Baghbaderani, S.S., Mokarian, P., Mohagheghian, F., Mohammed, L.J., Razmjou, A., 2021. Lithium bioleaching: an emerging approach for the recovery of Li from spent lithium ion batteries. Chemosphere 277, 130196. https://doi.org/10.1016/j.chemosphere.2021.130196.

Naseri, T., Bahaloo-Horeh, N., Mousavi, S.M., 2019a. Bacterial leaching as a green approach for typical metals recovery from end-of-life coin cells batteries. J. Clean. Prod. 220, 483–492. https://doi.org/10.1016/j.jclepro.2019.02.177.

Naseri, T., Bahaloo-Horeh, N., Mousavi, S.M., 2019b. Environmentally friendly recovery of valuable metals from spent coin cells through two-step bioleaching using Acidithiobacillus thiooxidans. J. Environ. Manag. 235, 357–367. https://doi.org/10.1016/j.jenvman.2019.01.086.

Niu, Z., Zou, Y., Xin, B., Chen, S., Liu, C., Li, Y., 2014. Process controls for improving bioleaching performance of both Li and Co from spent lithium ion batteries at high pulp density and its thermodynamics and kinetics exploration. Chemosphere 109, 92–98. https://doi.org/10.1016/j.chemosphere.2014.02.059.

Norambuena, J., Flores, R., Cárdenas, J.P., Quatrini, R., Chávez, R., Levicán, G., 2012. Thiol/Disulfide system plays a crucial role in redox protection in the acidophilic Iron-Oxidizing bacterium *Leptospirillum ferriphilum*. PLoS One 7, e44576. https://doi.org/10.1371/journal.pone.0044576.

Panda, S., Biswal, A., Mishra, S., Panda, P.K., Pradhan, N., Mohapatra, U., Sukla, L.B., Mishra, B.K., Akcil, A., 2015. Reductive dissolution by waste newspaper for enhanced meso-acidophilic bioleaching of copper from low grade chalcopyrite: a new concept of biohydrometallurgy. Hydrometallurgy 153, 98–105. https://doi.org/10.1016/j.hydromet.2015.02.006.

Rohwerder, T., Sand, W., 2007. Mechanisms and biochemical fundamentals of bacterial metal sulfide oxidation. In: Donati, E.R., Sand, W. (Eds.), Microbial Processing of Metal Sulfides. Springer, Dordrecht, Dordrecht, pp. 35–58.

Rohwerder, T., Gehrke, T., Kinzler, K., Sand, W., 2003. Bioleaching review part a: progress in bioleaching: fundamentals and mechanisms of bacterial metal sulfide oxidation. Appl. Microbiol. Biotechnol. 63, 239–248. https://doi.org/10.1007/s00253-003-1448-7.

Roy, J.J., Cao, B., Madhavi, S., 2021a. A review on the recycling of spent lithium-ion batteries (LIBs) by the bioleaching approach. Chemosphere 282, 130944. https://doi.org/10.1016/j.chemosphere.2021.130944.

Roy, J.J., Madhavi, S., Cao, B., 2021b. Metal extraction from spent lithium-ion batteries (LIBs) at high pulp density by environmentally friendly bioleaching process. J. Clean. Prod. 280, 124242 https://doi.org/10.1016/j.jclepro.2020.124242.

Roy, J.J., Rarotra, S., Krikstolaityte, V., Zhuoran, K.W., Cindy, Y.D.I., Tan, X.Y., Carboni, M., Meyer, D., Yan, Q., Srinivasan, M., 2022. Green recycling methods to treat lithium-ion batteries e-waste: a circular approach to sustainability. Adv. Mater. 34, 2103346 https://doi.org/10.1002/adma.202103346.

Ruhatiya, C., Gandra, R., Kondaiah, P., Manivas, K., Samhith, A., Gao, L., Lam, J.S.L., Garg, A., 2021. Intelligent optimization of bioleaching process for waste lithium-ion batteries: an application of support vector regression approach. Int. J. Energy Res. 45, 6152–6162. https://doi.org/10.1002/er.6238.

Sadeghabad, M.S., Bahaloo-Horeh, N., Mousavi, S.M., 2019. Using bacterial culture supernatant for extraction of manganese and zinc from waste alkaline button-cell batteries. Hydrometallurgy 188, 81–91. https://doi.org/10.1016/j.hydromet.2019.06.003.

Sand, W., Gehrke, T., 1999. Analysis and function of the EPS from the strong acidophile *Thiobacillus ferrooxidans*. In: Wingender, J., Neu, T.R., Flemming, H. (Eds.), Microbial Extracellular Polymeric Substances. Springer, Berlin, Heidelberg, Berlin, pp. 127–141.

Santos, F.M.F., Pina, P.S., Porcaro, R., Oliveira, V.A., Silva, C.A., Leão, V.A., 2010. The kinetics of zinc silicate leaching in sodium hydroxide. Hydrometallurgy 102, 43–49. https://doi.org/10.1016/j.hydromet.2010.01.010.

Seidel, A., Zimmels, Y., Armon, R., 2001. Mechanism of bioleaching of coal fly ash by *Thiobacillus thiooxidans*. Chem. Eng. J. 83, 123–130. https://doi.org/10.1016/S1385-8947(00)00256-4.

Sethurajan, M., Gaydardzhiev, S., 2021. Bioprocessing of spent lithium ion batteries for critical metals recovery – a review. Resour. Conserv. Recycl. 165, 105225 https://doi.org/10.1016/j.resconrec.2020.105225.

Shin, S.M., Kim, N.H., Sohn, J.S., Yang, D.H., Kim, Y.H., 2005. Development of a metal recovery process from Li-ion battery wastes. Hydrometallurgy 79, 172–181. https://doi.org/10.1016/j.hydromet.2005.06.004.

Silva, R.A., Park, J., Ilyas, S., Borja, D., Zhao, H., Urík, M., Rastegar, S.O., Kim, H., 2020. Biodegradation mechanism of arsenopyrite mine tailing with Acidithiobacillus ferrooxidans and influence of ferric supplements. Int. Biodeterior. Biodegrad. 153, 105042 https://doi.org/10.1016/j.ibiod.2020.105042.

Silverman, M.P., Ehrlich, H.L., 1964. Microbial formation and degradation of minerals. Adv. Appl. Microbiol. 6, 153–206. https://doi.org/10.1016/S0065-2164(08)70626-9.

Sun, L., Qiu, K., 2011. Vacuum pyrolysis and hydrometallurgical process for the recovery of valuable metals from spent lithium-ion batteries. J. Hazard Mater. 194, 378–384. https://doi.org/10.1016/j.jhazmat.2011.07.114.

Sun, L., Qiu, K., 2012. Organic oxalate as leachant and precipitant for the recovery of valuable metals from spent lithium-ion batteries. Waste Manage. (Tucson, Ariz.) 32, 1575–1582. https://doi.org/10.1016/j.wasman.2012.03.027.

Swain, B., 2017. Recovery and recycling of lithium: a review. Sep. Purif. Technol. 172, 388–403. https://doi.org/10.1016/j.seppur.2016.08.031.

Tanong, K., Coudert, L., Mercier, G., Blais, J., 2016. Recovery of metals from a mixture of various spent batteries by a hydrometallurgical process. J. Environ. Manag. 181, 95–107. https://doi.org/10.1016/j.jenvman.2016.05.084.

Tay, S.B., Natarajan, G., Rahim, M.N., bin, A., Tan, H.T., Chung, M.C.M., Ting, Y.P., Yew, W.S., 2013. Enhancing gold recovery from electronicwaste via lixiviant metabolic engineeringin *Chromobacterium violaceum*. Sci. Rep. 3, 2236. https://doi.org/10.1038/srep02236.

Tran, M.K., Rodrigues, M.F., Kato, K., Babu, G., Ajayan, P.M., 2019. Deep eutectic solvents for cathode recycling of Li-ion batteries. Nat. Energy 4, 339–345. https://doi.org/10.1038/s41560-019-0368-4.

Vaughan, D.J., 1984. Electronic structures of sulfides and leaching. In: Bautista, R.G. (Ed.), Hydrometallurgical Process Fundamentals. Plenum Press, New York, pp. 23–40.

Vera, M., Schippers, A., Sand, W., 2013. Progress in bioleaching: fundamentals and mechanisms of bacterial metal sulfide oxidation—part a. Appl. Microbiol. Biotechnol. 97, 7529–7541. https://doi.org/10.1007/s00253-013-4954-2.

Villares, M., Işıldar, A., Mendoza Beltran, A., Guinee, J., 2016. Applying an ex-ante life cycle perspective to metal recovery from e-waste using bioleaching. J. Clean. Prod. 129, 315–328. https://doi.org/10.1016/j.jclepro.2016.04.066.

Wang, J., Bai, J., Xu, J., Liang, B., 2009. Bioleaching of metals from printed wire boards by *Acidithiobacillus ferrooxidans* and *Acidithiobacillus thiooxidans* and their mixture. J. Hazard Mater. 172, 1100–1105. https://doi.org/10.1016/j.jhazmat.2009.07.102.

Wang, H., Zhang, X., Zhu, M., Tan, W., 2015. Effects of dissolved oxygen and carbon dioxide under oxygen-rich conditions on the biooxidation process of refractory gold concentrate and the microbial community. Miner. Eng. 80, 37–44. https://doi.org/10.1016/j.mineng.2015.06.016.

Wang, J., Tian, B., Bao, Y., Qian, C., Yang, Y., Niu, T., Xin, B., 2018. Functional exploration of extracellular polymeric substances (EPS) in the bioleaching of obsolete electric vehicle $LiNi_xCo_yMn_{1\text{-}x\text{-}y}O_2$ Li-ion batteries. J. Hazard Mater. 354, 250–257. https://doi.org/10.1016/j.jhazmat.2018.05.009.

Wang, J., Li, G., Yin, H., An, T., 2020. Bacterial response mechanism during biofilm growth on different metal material substrates: EPS characteristics, oxidative stress and molecular regulatory network analysis. Environ. Res. 185, 109451 https://doi.org/10.1016/j.envres.2020.109451.

Wang, J., Cui, Y., Chu, H., Tian, B., Li, H., Zhang, M., Xin, B., 2022. Enhanced metal bioleaching mechanisms of extracellular polymeric substance for obsolete $LiNi_xCo_yMn_{1\text{-}x\text{-}y}O_2$ at high pulp density. J. Environ. Manag. 318, 115429 https://doi.org/10.1016/j.jenvman.2022.115429.

Wu, W., Liu, X., Zhang, X., Li, X., Qiu, Y., Zhu, M., Tan, W., 2019. Mechanism underlying the bioleaching process of $LiCoO_2$ by sulfur-oxidizing and iron-oxidizing bacteria. J. Biosci. Bioeng. 128, 344–354. https://doi.org/10.1016/j.jbiosc.2019.03.007.

Wu, W., Li, X., Zhang, X., Gu, T., Qiu, Y., Zhu, M., Tan, W., 2020. Characteristics of oxidative stress and antioxidant defenses by a mixed culture of acidophilic bacteria in response to Co^{2+} exposure. Extremophiles 24, 485–499. https://doi.org/10.1007/s00792-020-01170-4.

Xin, B., Zhang, D., Zhang, X., Xia, Y., Wu, F., Chen, S., Li, L., 2009. Bioleaching mechanism of Co and Li from spent lithium-ion battery by the mixed culture of acidophilic sulfur-oxidizing and iron-oxidizing bacteria. Bioresour. Technol. 100, 6163–6169. https://doi.org/10.1016/j.biortech.2009.06.086.

Xin, Y., Guo, X., Chen, S., Wang, J., Wu, F., Xin, B., 2016. Bioleaching of valuable metals Li, Co, Ni and Mn from spent electric vehicle Li-ion batteries for the purpose of recovery. J. Clean. Prod. 116, 249–258. https://doi.org/10.1016/j.jclepro.2016.01.001.

Xu, Z., Liu, Z., Wang, S., Lu, Z., Zhang, Z., Wang, H., Jiang, F., 2022. Review on hydrometallurgical recovery of valuable metals from spent lithium-ion batteries. J. China Inst. Min. Technol. 51, 454–465. https://doi.org/10.13247/j.cnki.jcumt.001395.

Yan, K., Chen, Q., Xiong, Z., Wu, J., Zhang, Z., Xu, Z., Wang, R., Li, J., Zhong, S., 2022. A novel method for the recovery of li from spent Lithium-Ion batteries using reduction roasting–countercurrent leaching. Jom-Us 74, 3821–3832. https://doi.org/10.1007/s11837-022-05432-8.

Yang, J., Wang, Q., Wang, Q., Wu, T., 2008. Comparisons of One-Step and Two-Step bioleaching for heavy metals removed from municipal solid waste incineration fly ash. Environ. Eng. Sci. 25, 783–789. https://doi.org/10.1089/ees.2007.0211.

Yang, L., Lübeck, M., Lübeck, P.S., 2014. Deletion of glucose oxidase changes the pattern of organic acid production in *Aspergillus carbonarius*. Amb. Express 4, 54. https://doi.org/10.1186/s13568-014-0054-7.

Yang, X., Dong, P., Hao, T., Zhang, Y., Meng, Q., Li, Q., Zhou, S., 2020. A combined method of leaching and Co-Precipitation for recycling spent $Li_{0.6}Co_{0.2}Mn_{0.2}O_2$ cathode materials: process optimization and performance aspects. Jom-Us. 72, 3843–3852. https://doi.org/10.1007/s11837-020-04263-9.

Yang, H., Li, J., Tong, L., Ma, P., Zhang, Q., Jin, Z., 2021. Role of hydrolyzed rice husk in pyrite bio-oxidation. Geochemistry 81, 125775. https://doi.org/10.1016/j.chemer.2021.125775.

Yang, C., Zhang, J., Liang, G., Jin, H., Chen, Y., Wang, C., 2022. An advanced strategy of "metallurgy before sorting" for recycling spent entire ternary lithium-ion batteries. J. Clean. Prod. 361, 132268 https://doi.org/10.1016/j.jclepro.2022.132268.

Yao, Y., Zhu, M., Zhao, Z., Tong, B., Fan, Y., Hua, Z., 2018. Hydrometallurgical processes for recycling spent Lithium-Ion batteries: a critical review. Acs Sustain. Chem. Eng. 6, 13611–13627. https://doi.org/10.1021/acssuschemeng.8b03545.

Yao, L.P., Zeng, Q., Qi, T., Li, J., 2020. An environmentally friendly discharge technology to pretreat spent lithium-ion batteries. J. Clean. Prod. 245, 118820 https://doi.org/10.1016/j.jclepro.2019.118820.

Yu, J., Liu, Y., Han, S., Tan, Q., Liu, L., Li, J., 2021a. Unveiling sodium ion pollution in Spray-Dried precursors and its implications for the green upcycling of spent Lithium-Ion batteries. Environ. Sci. Technol. 55, 14897–14905. https://doi.org/10.1021/acs.est.1c05511.

Yu, J., Lin, M., Tan, Q., Li, J., 2021b. High-value utilization of graphite electrodes in spent lithium-ion batteries: from 3D waste graphite to 2D graphene oxide. J. Hazard Mater. 401, 123715 https://doi.org/10.1016/j.jhazmat.2020.123715.

Yu, J., Zou, S., Xu, G., Liu, L., Zhao, M., Li, J., 2022. In-situ enhanced catalytic reforming behavior of cobalt-based materials with inherent zero-valent aluminum in spent lithium ion batteries. Appl. Catal. B Environ. 303, 120920 https://doi.org/10.1016/j.apcatb.2021.120920.

Yu, J., Li, J., Zhang, S., Wei, F., Liu, Y., Li, J., 2023. Mechanochemical upcycling of spent $LiCoO_2$ to new $LiNi_{0.80}Co_{0.15}Al_{0.05}O_2$ battery: an atom economy strategy. Proc. Natl. Acad. Sci. U. S. A. 120 https://doi.org/10.1073/pnas.2217698120.

Zeng, G., Deng, X., Luo, S., Luo, X., Zou, J., 2012. A copper-catalyzed bioleaching process for enhancement of cobalt dissolution from spent lithium-ion batteries. J. Hazard Mater. 199–200, 164–169. https://doi.org/10.1016/j.jhazmat.2011.10.063.

Zeng, G., Luo, S., Deng, X., Li, L., Au, C., 2013. Influence of silver ions on bioleaching of cobalt from spent lithium batteries. Miner. Eng. 49, 40–44. https://doi.org/10.1016/j.mineng.2013.04.021.

Zeng, G., Luo, S., Li, H., Wang, X., 2014. Innovative electrochemistry technique for leaching cobalt from spent lithium ion batteries. Asian J. Chem. 26, 1099–1102. https://doi.org/10.14233/ajchem.2014.15933.

Zhang, Y., Meng, Q., Dong, P., Duan, J., Lin, Y., 2018. Use of grape seed as reductant for leaching of cobalt from spent lithium-ion batteries. J. Ind. Eng. Chem. 66, 86–93. https://doi.org/10.1016/j.jiec.2018.05.004.

Zhang, G., Yuan, X., He, Y., Wang, H., Zhang, T., Xie, W., 2021. Recent advances in pretreating technology for recycling valuable metals from spent lithium-ion batteries. J. Hazard Mater. 406, 124332 https://doi.org/10.1016/j.jhazmat.2020.124332.

Zhao, Y., Yuan, X., Jiang, L., Wen, J., Wang, H., Guan, R., Zhang, J., Zeng, G., 2020. Regeneration and reutilization of cathode materials from spent lithium-ion batteries. Chem. Eng. J. 383, 123089 https://doi.org/10.1016/j.cej.2019.123089.

Zhao, S., Wang, G., Yang, H., Chen, G., Qiu, X., 2021. Agglomeration-aggregation and leaching properties of mechanically activated chalcopyrite. T Nonferr. Metal. Soc. 31, 1465–1474. https://doi.org/10.1016/S1003-6326(21)65590-5.

Zheng, R., Wang, W., Dai, Y., Ma, Q., Liu, Y., Mu, D., Li, R., Ren, J., Dai, C., 2017. A closed-loop process for recycling $LiNi_xCo_yMn_{(1\text{-}x\text{-}y)}O_2$ from mixed cathode materials of lithium-ion batteries. Green Energy Environ. 2, 42–50. https://doi.org/10.1016/j.gee.2016.11.010.

Zheng, X., Zhu, Z., Lin, X., Zhang, Y., He, Y., Cao, H., Sun, Z., 2018a. A mini-review on metal recycling from spent lithium ion batteries. Eng. Plast. 4, 361–370. https://doi.org/10.1016/j.eng.2018.05.018.

Zheng, C., Zhang, L., Chen, M., Zhao, X.Q., Duan, Y., Meng, Y., Zhang, X., Shen, R.F., 2018b. Effects of cadmium exposure on expression of glutathione synthetase system genes in *Acidithiobacillus ferrooxidans*. Extremophiles 22, 895–902. https://doi.org/10.1007/s00792-018-1046-3.

Zhou, S., Zhang, Y., Meng, Q., Dong, P., Yang, X., Liu, P., Li, Q., Fei, Z., 2021. Recycling of spent $LiCoO_2$ materials by electrolytic leaching of cathode electrode plate. J. Environ. Chem. Eng. 9, 104789 https://doi.org/10.1016/j.jece.2020.104789.